

Integrated Heteroepitaxial Photodetectors

by

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Abstract

Optical detection in the near-infrared and telecommunication bands has historically been performed using single-crystal bulk Ge, but the development of Ge-on-Si epitaxy reduced fabrication costs and opened doors for usage in applications including optical communications and infrared imaging. To reap the benefits of monolithic integration and incorporation in the back-end-of-line (BEOL) stack, low processing temperatures ($< 450^\circ$) are required. Using novel processing methods and strategic anneals, we have demonstrated that low temperature Ge growths on silicon can achieve low defect densities required for high performance. In this work, Ge-on-Si p-i-n photodetectors illuminated under normal incidence have demonstrated comparable responsivity and dark current density to devices processed at high temperatures. Relatively low temperature anneals (500°C) increased performance, but as-grown diodes also showed a responsivity of 0.11 A/W and $J_d = 340\text{ }\mu\text{A/cm}^2$. Annealing conditions of 500°C 3 hr improved such performance to 0.15 A/W and $J_d = 170\text{ }\mu\text{A/cm}^2$.

In the mid-wave infrared (MWIR), photodetection has been successfully implemented for decades using the II-VI material set, $\text{Hg}_{1-x}\text{Cd}_x\text{Te}$. Extensive research pushed HgCdTe to nearly reach its theoretical performance limit, while also highlighting its inherent shortcomings for commercialization. An upcoming material set, $\text{InAs}_x\text{Sb}_{1-x}$ has the potential to overcome such barriers while promising comparable performance. In this work, Lumerical simulations were performed to optimize a waveguide-integrated photodetector that incorporated an $\text{InAs}_{0.85}\text{Sb}_{0.15}$ homojunction and was straightforward to fabricate, assuming successful epitaxy growths. The photodetector design promoted 30% light absorption after $20\text{ }\mu\text{m}$ propagation into the detection region.

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Chapter 1

Near-IR/Back-end-of-line compatible Ge-on-Si

1.1 Introduction

The current state of information technology relies on ultra large-scale integrated electronic circuits, which have progressed exponentially in communication capacity and computation performance. However, electronic circuits face inherent limitations in wire resistance, line capacitance, electromagnetic interference and skin effects, which prevent this existing platform from meeting future energy consumption and bandwidth requirements. Power efficiency improvements of 1000x are required over the next two decades, yet the existing architecture requires 200x more energy to transport a bit to a nearest-neighbor chip than to simply operate on it. Photonic integration is a highly attractive solution for overcoming these limitations with relatively distance independent operation, and high bandwidth potential without RC delays that can be improved even further with the incorporation of wavelength-division multiplexing (WDM) within a single fiber [1].

Silicon-on-insulator (SOI) substrates have so far been the most commercialized platform for monolithic electronic-photonic integration, but the SOI CMOS processes are cost-prohibitive for high volume manufacturing. Existing bulk CMOS processes are not compatible with direct photonic integration, but a proposed photonic plat-

form compatible with bulk CMOS processes incorporated polycrystalline Si on islands of SiO_2 in the front-end-of-line (FEOL) [2]. A full photonic library of passive and active components, as well as high-bandwidth optical transceivers, were fabricated at the CMOS foundry in SUNY Albany and tested for performance, alongside the fabrication of transistors to demonstrate successful integration with nanoelectronics. This photonic platform overcame cost limitations with photonic integration, but did not address the large footprint required for incorporation of photonic devices in the FEOL.

An alternative to FEOL integration is back-end-of-line integration (BEOL), which would allow for the large footprint of the photonic components to be integrated vertically as opposed to horizontally as shown in Figure 1-1. A key issue with BEOL integration is the lack of semiconductor material in the dielectric stack; however, it has been shown that Ge single crystals may be grown on amorphous substrate. In McComber’s work, Ge was grown on amorphous Si via confinement in 1D and 2D channels. Such geometrically-confined lateral growth (GCLG) promoted the fastest grain growth for high surface energy crystal planes, which ultimately defined the dominant orientation for the film and promoted single-crystal growth [3].

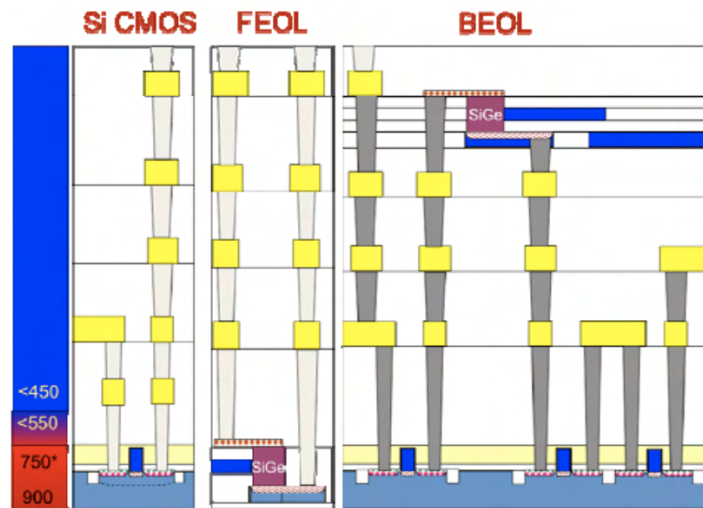


Figure 1-1: Monolithic photonic integration in Si CMOS processes [1].

Successful results of single crystal growth on amorphous substrate opened a path forward for BEOL integration, motivating this thesis work on BEOL-compatible Ge

on Si photodetectors. One unresolved issue is the 4% lattice mismatch between Ge and Si, which causes misfit dislocations at the interface and threading dislocations throughout the Ge film, and has previously been resolved with high temperature cyclic thermal annealing [4]. Direct wafer bonding has also been explored for high quality integration without high temperature exposure of Si-CMOS [5]. In this work, high performance photodetectors were fabricated under the restraints of the BEOL thermal budget, and performance improvements after low temperature anneals of 450-550 °C were also evaluated. These devices were designed for normal incidence light and could be applied directly in LIDAR applications, but the same principles could also be applied for waveguide-integrated photodetectors.

1.2 Device Design

The goal of device design and fabrication was not only to showcase high performance 1310 nm Ge photodetection, but also to demonstrate that the Ge epi layers grown at $< 450^\circ$ were of high quality and low defect density. Device performance from annealed wafers could then be used to correlate and model defect removal kinetics with annealing conditions. Further study could include the direct evaluation of defects via transmission electron microscopy (TEM) or etch-pit density counting. A verification of surface roughness, which forms from islanding during the growth process and causes integration issues, would be evaluated with atomic force microscopy (AFM) [6]. In this work, we implemented a design, demonstrated high performance photodetection, and estimated the minority carrier lifetime from dark current density.

In a photodiode, current has an exponential dependence with voltage, such that applying forward bias allows current to pass while reverse bias inhibits current flow. Any remaining current at reverse bias is commonly referred to as dark current, leakage current, or saturation current I_s , and its dependence on voltage is shown in [Equation 1.1](#). The above scenario assumes no incident light.

$$I = I_s (e^{\frac{qV}{kT}} - 1) \quad (1.1)$$

A design optimization would focus on reducing dark current while also ensuring efficient carrier collection to maximize photocurrent. The heterojunction designs selected in this work are shown in [Table 1.1](#), where the Ge epi sits above the pn junction in Si. The notation "Gen3" indicates that the Ge doping was graded, while "Gen2" denotes homogenous doping. All of the Gen2 800°C 2 min wafers had a low doping level of $2 \times 10^{16} \text{ cm}^{-3}$ and were annealed at high temperature, while the rest of the Gen2 wafers had a higher doping level of $5 \times 10^{17} \text{ cm}^{-3}$, arrived as-grown and were selectively exposed to low temperature anneals between 450-550 °C in the ultra high vacuum chemical vapor deposition (UHVCVD) chamber. An alternative design could include a homojunction instead of a heterojunction, where p-Ge would be grown directly on n-Si substrate. However, the heterojunction design had the

advantage of being able to separately tune photon absorption and carrier collection. Furthermore, a key advantage in the Gen3 graded doping design was that the built-in E field within the p-Ge would generate its own drift current and limit surface and sidewall recombination, maximizing quantum efficiency and reducing transit time.

Table 1.1: Ge-on-Si heterojunction designs used in device fabrication, including growth and annealing conditions.

Layer	Type	Dopant	Thickness (μm)	Carrier Concentration (cm^{-3})		
				Gen2 800C 2min	Gen2 + annealing	Gen3 + annealing
p-Ge	Epi	B	0.5	$2*10^{16} \text{ cm}^{-3}$	$5*10^{17} \text{ cm}^{-3}$	$10^{17}\text{-}10^{18}\text{-}10^{19} \text{ cm}^{-3}$
p-Si	Epi	B	1.0	$1*10^{15}$	$1*10^{15}$	$1*10^{15}$
n-Si	Substrate	Sb	1.0	$1.5*10^{18}$	$1.5*10^{18}$	$1.5*10^{18}$
Growth Temperature of Ge epi				$< 450^\circ\text{C}$	$< 450^\circ\text{C}$	$< 450^\circ\text{C}$
Annealing Conditions				800°C 2 min	As-grown, or $450\text{-}550^\circ\text{C}$	As-grown, or $450\text{-}550^\circ\text{C}$

The layer thicknesses and doping levels greatly impact the device characteristics. Because Ge has a smaller bandgap than Si (0.67 eV vs 1.14 eV) and a similar electron affinity (4.0 eV vs 4.05 eV), a natural energy barrier to electron flow is created when the Fermi levels align. Since the Fermi level is the level at which the probability of electron occupancy is one-half, electron diffusion naturally occurs from p-Ge to p-Si. Similarly, electrons also diffuse from n-Si to p-Si, creating the built in junction for drift current, also referred to as the depletion zone or space charge region (SCR). In reverse bias, a negative voltage is applied to the p-Ge layer, widening the SCR.

Carrier movement from the p-Ge absorption region to p-Si is obviously hindered with the energy barrier. However, a thinner p-Si layer allows the depleted region to reach the p-Ge interface at relatively low reverse bias V_r , effectively removing the barrier and enabling carrier movement. [Equation 1.2](#) and [Equation 1.3](#) quantitatively show how doping levels also impact carrier movement at a given voltage [7]. The built-in voltage ϕ_{bi} increases with the natural log of doping concentration, such that higher doping on either side of the p-Ge/p-Si junction will require a higher V_r to overcome it. It should be noted that [Equation 1.2](#) is intended for pn junctions within a single material, and applying it to a heterointerface is not precisely correct; nonetheless, the trend is the same.

$$\phi_{bi} = \frac{kT}{q} \ln \frac{N_D N_A}{n_i^2} \quad (1.2)$$

$$x_p = \sqrt{\frac{2\epsilon N_D \phi_{bi}}{q N_A (N_D + N_A)}} \quad (1.3)$$

Also of importance is the impact of doping on the edge position of the SCR. As previously mentioned, the edge of the SCR should reach as far into the p-Si region as possible, in order to deplete the p-Si region at a relatively low V_r . From Equation 1.3, elevated doping of p-Si would decrease x_p and increase the onset voltage. Another observation is that as long as N_D remains much higher than N_A , there is minimal impact of N_D on device performance. However, if N_D drops below N_A , then x_p will decrease and the edge of the SCR will shift. The built-in voltage will also drop, and the resulting onset voltage is a combination of these two effects.

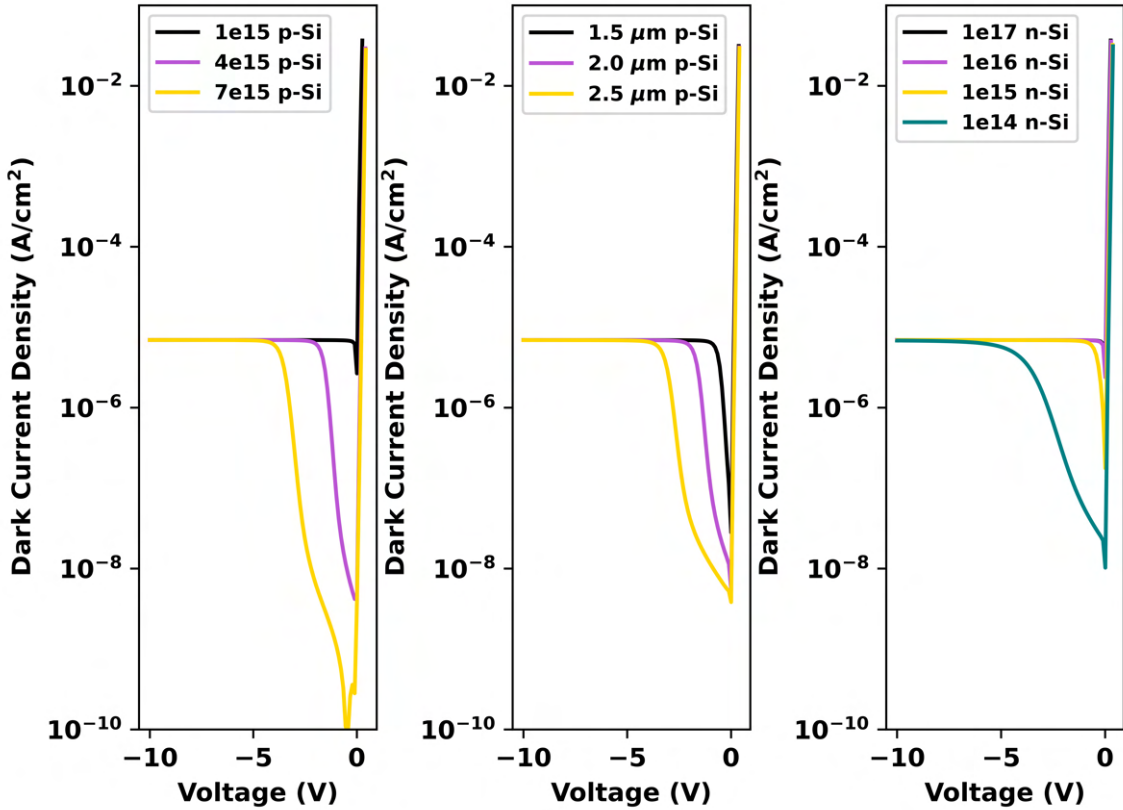


Figure 1-2: PC1D simulations of Gen3 device structure, evaluating the impact of n-Si and p-Si doping/thickness on J_d and onset voltage.

These intuitive observations translated well to PC1D simulation results shown in [Figure 1-2](#). The simulation assumed a one-dimensional problem, with no surface recombination velocity at the sidewall, top or bottom surfaces. Each simulation was based on the Gen3 structure from [Table 1.1](#), with a minority carrier lifetime of $\tau_n = 1 \text{ ns}$ in p-Ge. As expected, the onset voltage increased steadily with p-Si doping or thickness, while the n-Si doping level showed much higher design tolerance. Only when the n-Si doping fell below that of p-Si did the onset voltage increase, which was well explained by [Equation 1.3](#). Thus, we had a design that was sensitive to fabrication error in p-Si, but less sensitive to error in n-Si. Drawing an additional connection to theory, the dark current density, J_d at low voltages was observed to drop with increased p-Si doping at 4×10^{15} and $7 \times 10^{15} \text{ cm}^{-3}$. [Equation 1.2](#) confirmed that such elevated doping also increased the energy barrier between p-Ge and p-Si, thereby causing a drop in J_d at low voltages. Increased p-Si thickness or low doping of n-Si simply caused the edge of the SCR region in p-Si to shift, increasing the onset voltage without causing a drop in J_d prior to onset.

The saturation current density, J_s remained constant at $6.9 \text{ } \mu\text{A}/\text{cm}^2$ regardless of the design changes. Since the minority carrier lifetime was kept at $\tau_n = 1 \text{ ns}$ in p-Ge, this result was expected given that essentially all of the J_s contribution was derived from Ge. The quantum efficiency, or the ratio of carriers collected to incident photons, was also the same in nearly every scenario at 40-41% post-voltage onset. In the PC1D simulation, the incident intensity was set to $0.1 \text{ W}/\text{cm}^2$ at 1310 nm, and the surface was assumed to have a perfect AR coating. An interesting limitation of the simulation software, the absorption of light within the Ge layer was homogeneous with depth rather than dropping with $e^{-\alpha x}$, where α = absorption coefficient. Critical for maximizing photocurrent in the detector, quantum efficiency stayed the same regardless of design, since all simulation points were taken after the energy barrier was reduced. An example of this energy barrier reduction is provided in [Figure 1-3](#) for the Gen3 $2.5 \text{ } \mu\text{m}$ p-Si design scenario, and as shown the high onset voltage of 6 V was enough to lower the barrier and promote drift current in the p-Si region. The carrier lifetime τ_n within the Ge layer would have the greatest impact on J_s and quantum

efficiency, and in fact at $\tau_n = 10 \mu\text{s}$, $J_s = 0.16 \mu\text{A}/\text{cm}^2$ and quantum efficiency was slightly higher than before at 42.8%.

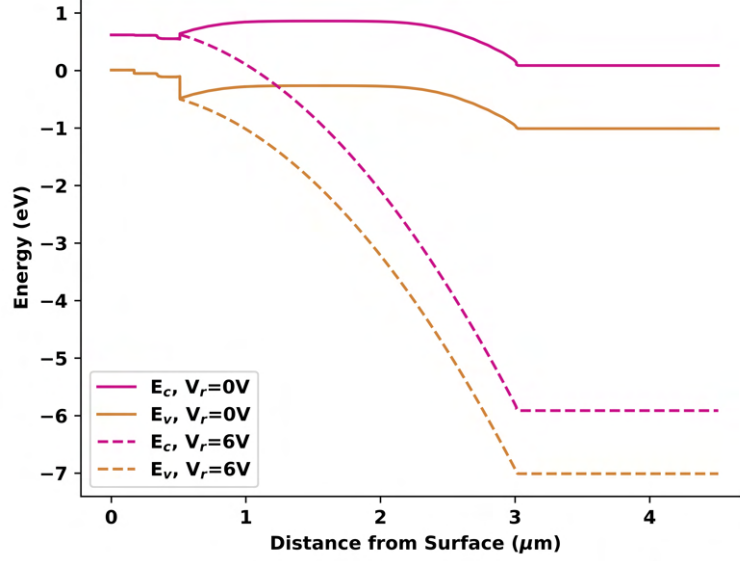


Figure 1-3: PC1D energy band diagrams for Gen3 device structure with $2.5 \mu\text{m}$ p-Si.

The p-Ge layer thickness was designed to absorb a significant portion of incident light to maximize quantum efficiency, while also accommodating the short diffusion length that could prevent photogenerated carriers from reaching the E field in p-Si. Equation 1.4 relates mobility to diffusivity, and Equation 1.5 relates the diffusivity to diffusion length [7]. From Hall Effect, a hole mobility for low-temperature p-Ge was measured and converted to an approximate electron mobility using a ratio of conductivity electron/hole effective masses. With $\mu_e = 100 \frac{\text{cm}^2}{\text{V}\cdot\text{s}}$, and carrier lifetime $\tau_e = 1 \text{ ns}$, the diffusion length in p-Ge was approximately $0.5 \mu\text{m}$. Thus, the design of $0.5 \mu\text{m}$ p-Ge successfully accounts for some limitations in diffusion length from material quality.

$$D_e = \mu_e \frac{kT}{q} \quad (1.4)$$

$$L = \sqrt{D_e \tau_e} \quad (1.5)$$

In this work, a low-doped n-Si region did in fact form unexpectedly during fabrication as a result of Sb diffusion from the highly doped n+ substrate into p-Si. Since p-Si

epi was grown at high enough temperatures to promote significant donor diffusion, the supplier attempted to perform a p-type counter-doping at the beginning of the p-Si epitaxial growth. However, this counter-doping was not performed consistently and it lead to wide variations in doping profiles at the p-Si/n-Si interface, as seen in the spreading resistance profiles (SRP) in [Figure 1-4](#). The lack of counter-doping in the second wafer lot, containing Gen3 as-grown resulted in a transition region of approximately 1 μm within the n-Si region, dipping far below the p-Si doping level of 1×10^{15} , which as previously mentioned would cause an increase in the onset voltage.

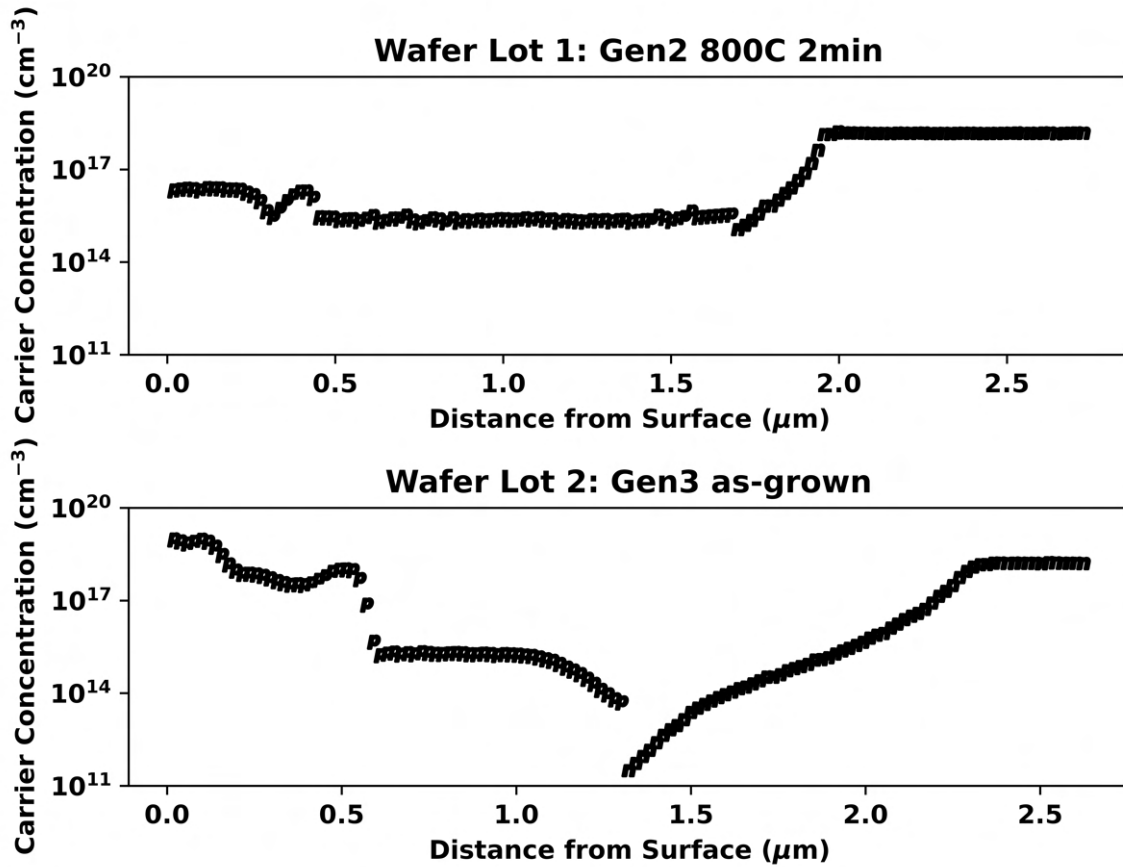


Figure 1-4: Spreading resistance profiles (SRP) for Gen2 800°C 2 min and Gen3 as-grown wafers sourced from two separate lots.

Analysis of SRP profiles is not entirely straightforward. Although [Figure 1-4](#) appears to show two doping profiles, the SRP method is based on resistivity measurements with depth taken from a beveled cross-section, which is then converted to carrier concentration from published carrier mobility values. From an experimental

standpoint, carrier concentration is being measured from the top-down, and at any given depth the carrier concentration is dependent only on the layers and carrier redistribution beneath it. Thick or highly-doped layers translate well to carrier concentration, and can be directly interpreted as dopant concentration. However, thinner layers, especially with lower doping are highly susceptible to space-charge effects, widely referred to as carrier spilling in SRP [8], and the SRP method can therefore provide misleading results. The p-Ge and n-Si layers in [Figure 1-4](#) may therefore be interpreted as approximate dopant profiles, but the p-Si layer is too affected by carrier redistribution from the p-Si/n-Si junction to be analyzed directly. More rigorous analysis may be performed with a Poisson solver to extract a dopant profile. However, even without such analysis the large transition region in n-Si is clearly visible in Wafer Lot 2, and the p-Ge layers in both wafer lots have some abnormalities and unexpected peaks from the profiles that were requested from [Table 1.1](#).

The abnormalities in the SRP profiles were observable in the experimental results. [Figure 1-5](#) shows the dark I-V curves for two diodes, one from each wafer lot. In the Wafer Lot 1, Gen2 800°C 2 min had a voltage onset of $V_r = 2\text{ V}$, due perhaps to the slightly thicker p-Si region when simulation predicted that the voltage onset would be much closer to zero. From Wafer Lot 2, a Gen3 sample was annealed at 500 °C for 3 hr and had a much higher voltage onset closer to $V_r = 4\text{ V}$. The Sb out-diffusion from the n+ substrate likely caused the significantly higher onset voltage. J_d was orders of magnitude higher for Gen3 500°C 3 hr, since the material quality was not as high for a lower temperature anneal.

[Figure 1-6](#) shows the final device structure for Gen3, taking the transition region into account. Given the results from SRP, the graded doping concentrations are approximated. The diode contains two top contacts (Ti/TiN/Al) as well as ALD passivation to limit surface and sidewall recombination. Detailed information on fabrication is given in the following section.

The primary contribution to dark current is from the Ge layer, but a direct correlation between J_s and τ_n has not been made. [Equation 1.6](#) defines the saturation current density for a pn junction within a given material, resulting from minority car-

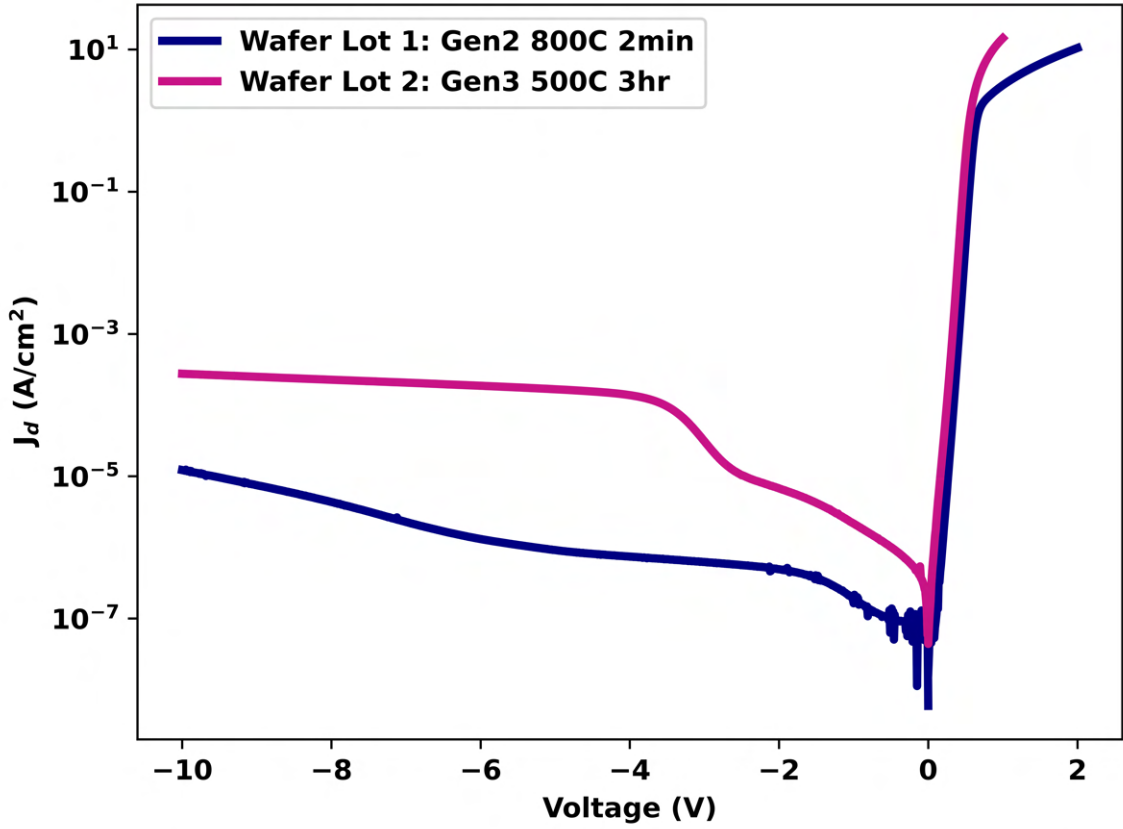


Figure 1-5: Dark I-V characteristics for Gen2 800°C 2 min and Gen3 500°C 3h r wafers sourced from separate lots.

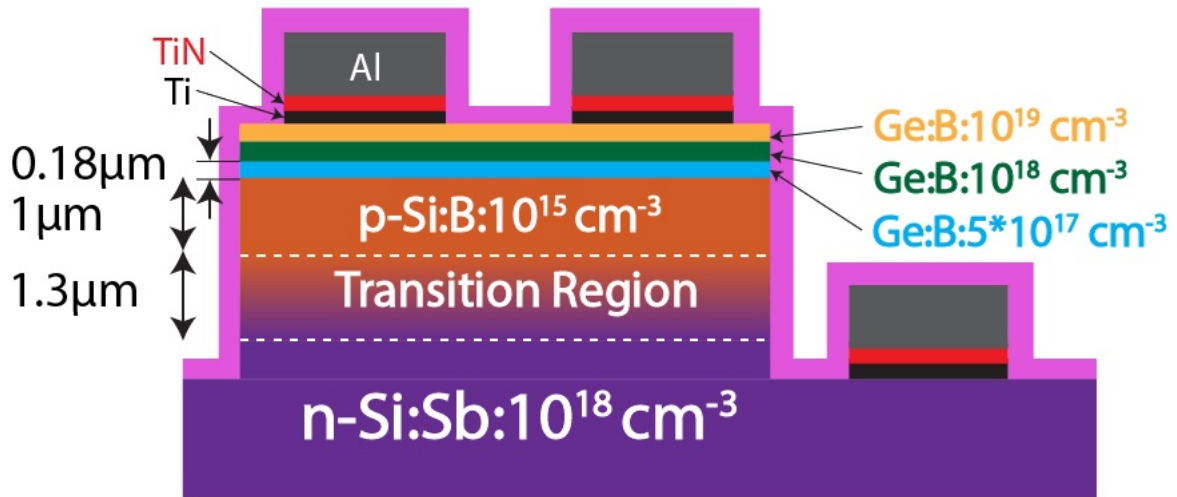


Figure 1-6: Gen3 diode cross section, with approximated device structure based on SRP measurements [9].

rier extraction [7]. Although the heterojunction device design does not fit perfectly

within this model, the model contains an inverse square-root relation with carrier lifetime that can be used to give a prediction for J_s and vice versa.

$$J_s = qN_cN_v \exp \frac{-E_g}{kT} \left(\frac{1}{N_A} \sqrt{\frac{D_e}{\tau_e}} + \frac{1}{N_D} \sqrt{\frac{D_h}{\tau_h}} \right) \quad (1.6)$$

Figure 1-7 shows the relation between saturation current density J_s and carrier lifetime with temperature dependence, for a theoretical Ge homojunction where $N_A = N_D = 5 \cdot 10^{17} \text{ cm}^{-3}$. As expected, J_s increases significantly as carrier lifetime decreases, with $J_s = 0.1 \mu\text{A}/\text{cm}^2$ at $\tau = 10 \mu\text{s}$ and $J_s = 3.4 \mu\text{A}/\text{cm}^2$ at $\tau = 10 \text{ ns}$. This is extremely similar to results from PC1D that predicted $J_s = 0.2 \mu\text{A}/\text{cm}^2$ at $\tau = 10 \mu\text{s}$ and $J_s = 3.8 \mu\text{A}/\text{cm}^2$ at $\tau = 10 \text{ ns}$. The above values were reported at 300 K, but a temperature increase to 450 K causes J_s to increase by roughly five orders of magnitude. Most of the terms in Equation 1.6 have temperature dependence, but this notable increase is primarily due to intrinsic carrier concentration. Bandgap also decreases linearly with temperature, dropping from 0.66 eV to 0.60 eV over a 150 K interval.

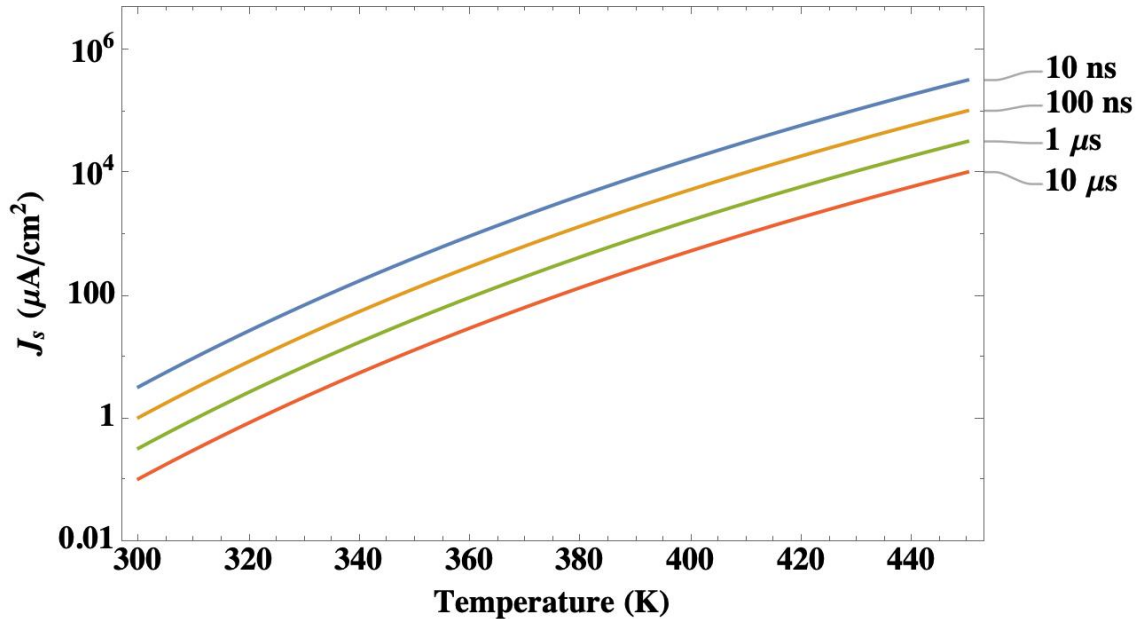


Figure 1-7: Effect of temperature and carrier lifetime on J_s , based on a theoretical model of p-Ge/n-Ge junction with $5 \cdot 10^{17} \text{ cm}^{-3}$ homogeneous doping.

Equation 1.6 also provides some illumination as to why the simulated J_s is slightly

lower for Gen2 than Gen3, at $2.2 \mu\text{A}/\text{cm}^2$ and $6.9 \mu\text{A}/\text{cm}^2$, respectively. The design of a built-in E field for Gen3 essentially creates additional pn junctions in Ge, which each have dark current contributions. However, the result is within the same order of magnitude, and in experiment similar results would be expected for similar material quality. Based on results at $V_r = 4 \text{ V}$ in [Figure 1-5](#), the estimated minority carrier lifetime is roughly 200 ns for low-temperature grown Ge annealed at 800°C , and 0.005 ns for Ge annealed at 500°C for 3 hr. This result is an initial approximation - a more complex model would incorporate the heterojunction design, estimate trap-assisted generation in p-Ge due to defects and assume collection of all carriers.

1.3 Device Fabrication and I-V Characteristics

1.3.1 Fabrication

To fabricate Ge-on-Si photodiodes, wafers were sourced from Lawrence Semiconductor Research Laboratory. For annealing, the wafers were loaded under vacuum in the UHVCVD chamber in SEL and heated for a set time and temperature in the tube furnace. All subsequent fabrication was performed in MIT.Nano.

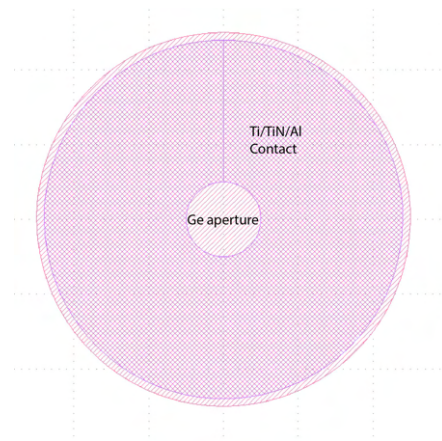


Figure 1-8: Top view of diode with $250 \mu\text{m}$ radius.

[Figure 1-6](#) and [Figure 1-8](#) show the cross section and top view for the diodes, and [Table 1.2](#) shows the process sequence. The Ge-on-Si wafers were initially cleaved and sonicated in acetone, methanol and IPA. Negative photoresist nLOF 2035 is then spun

Table 1.2: Process flow for Ge-on-Si photodiodes.

Step	Lab	Machine	Action
1	SEL	UHV/CVD	Anneal under vacuum, 450 °C - 550 °C
2	Nano	Cleaving station	Cleave wafer into pieces
3	Nano	Solvent-L06	Sonicate for 3 min each in acetone, methanol and IPA
4	Nano	Spinner-Resist-U12	Spin coat 4-5 μm nLOF 2035
5	Nano	Hotplate-Tower-U12	Softbake
6	Nano	DirectWrite-MLA150	Expose patterning
7	Nano	Develop-U12	Develop photoresist
8	Nano	RIE-Cl-Samco-230iP / RIE-Mixed-Samco-230iP	Dry etching of pattern transfer to make Ge on p-Si cylinders using Cl_2
9	Nano	Dektak-XT	Check etch depth
10	Nano	Asher-Barrel-Thierry / Asher-ESI	Photoresist strip
11	Nano	Spinner-PMGI-U12	PMGI layer
12	Nano	Spinner-Resist-U12	AZ3312 layer
13	Nano	Hotplate-Tower-U12	Softbake
14	Nano	DirectWrite-MLA150	Expose patterning
15	Nano	Develop-U12	Develop photoresist
16	Nano	Asher-Barrel-Thierry	Descum
17	Nano	E-beam AJA	Deposit 20 nm Ti/20 nm TiN/300 nm Al to form top Si and top Ge contacts
18	Nano	Liftoff-L08	NMP with short sonication
19	Nano	Asher-Barrel-Thierry/Asher-ESI	Resist removal
20	Nano	RTA-1300C-ASOne150-5Gas	Optional, 450 °C 30 s in N_2
21	Nano	ALD-Ozone	Deposit 20 nm Al_2O_3 to passivate sidewalls

at 1000 rpm, followed by a 90 s softbake at 110 °C, 140 mJ/cm² exposure at 375 nm in the direct-write MLA tool and development in AZ MIF 300 for 90-120 s with some agitation. Dry etching was then performed in the RIE-Cl₂-Samco using Cl₂ plasma at room temperature, followed by oxygen ashing to remove the remaining photoresist.

Metal contacts (Ti/TiN/Al) were patterned using PMGI and positive resist AZ3312. PMGI SF5 was spin-coated at 3500 rpm for 30 s, and softbaked for 60 s @ 235 °C. After repeating this PMGI coat three times to ensure a thick enough undercut layer for liftoff, AZ3312 was spun @ 3500 rpm for 30 s and exposed at 160 mJ/cm² using the 375 nm laser. Development time was approximately 90-110 s in AZ300 MIF with some agitation. Samples were then loaded in the Ebeam AJA until the baseline pressure reached at least 2.0e-6 torr, and 20 nm Ti/20 nm TiN/300 nm Al were then deposited. The samples were stationary during deposition as the Ebeam AJA does not employ any planetary rotation. Liftoff in NMP took 2 hours followed by some sonication in the NMP solution, a quick sonication in acetone and IPA rinse. Any residual photoresist at this point (if any) was removed via descum in an oxygen asher prior to

ALD, both of which had a thermal budget of 250 °C. To make the ohmic contact with the n-Si, a rapid thermal anneal (RTA) was performed at 450 °C for 30 s in N₂ environment for some diode lots, with an overshoot temperature of approximately 465 °C. For clarity, the results reported in this work did not incorporate the RTA step. In the ALD-Ozone, 20 nm of ALD was deposited using H₂O and TMAH.

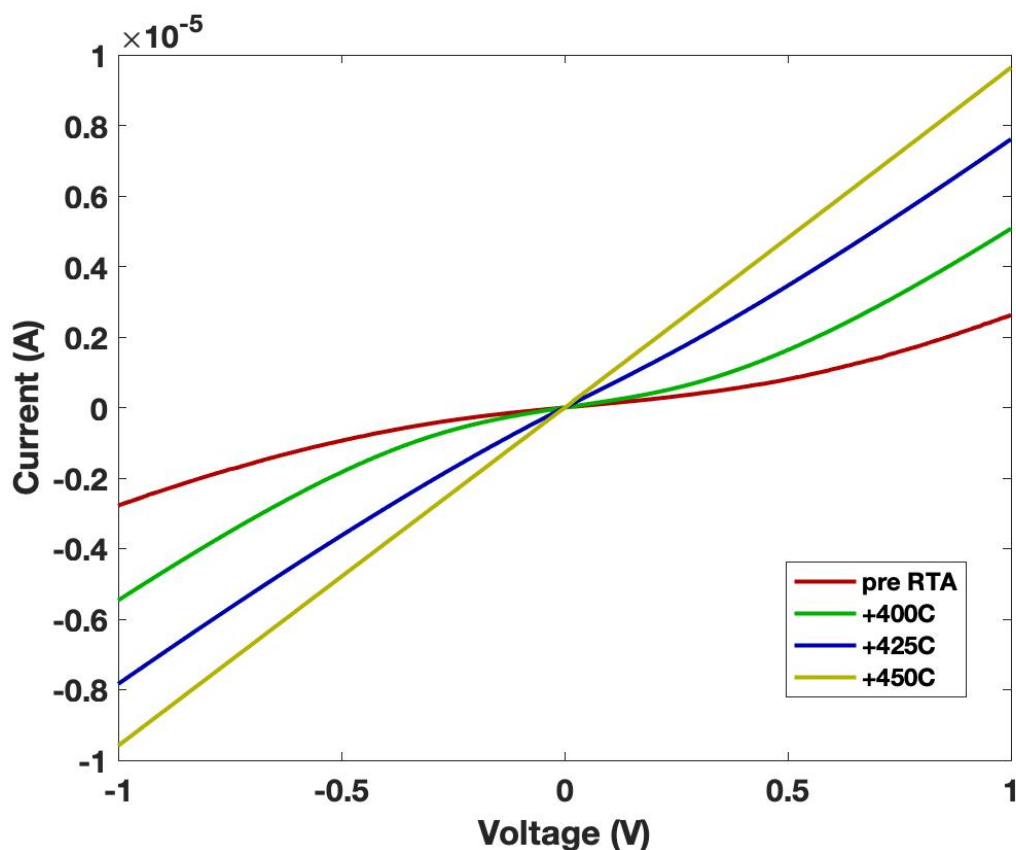


Figure 1-9: Ohmicity of Ti/TiN/Al contacts on n-Si with RTA temperature.

The RTA process temperature was determined after testing the ohmicity of multiple contacts at a series of temperatures ranging from 400-450 °C. The RTA contained an SiC-coated graphite susceptor and susceptor lid, with heat lamps placed above the susceptor lid and a pyrometer situated to view the temperature from the bottom of the susceptor. This setup was designed to provide better heating uniformity within the sample. The gap between the n-Si metal contacts was 1000 μ m, and unannealed contacts prior to RTA demonstrated a non-linear dependence with voltage as shown in [Figure 1-9](#), indicative of two back-to-back Schottky contacts. Subsequent anneals

performed on the same sample gradually improved ohmic characteristics, demonstrating linearity at 450 °C. The mechanism for Ti-based ohmic contacts on n+-Si is a solid-state reaction that forms an amorphous interlayer with low contact resistance between 450-600 °C [10].

In the diode mask design, five different diode sizes were included: radii 250 μm , 500 μm , 750 μm , 1000 μm , and 2000 μm . Annealing temperatures and times ranged from 450 °C to 550 °C, and one to nine hours. See Figure 1-10 for an example of a typical diode mask.

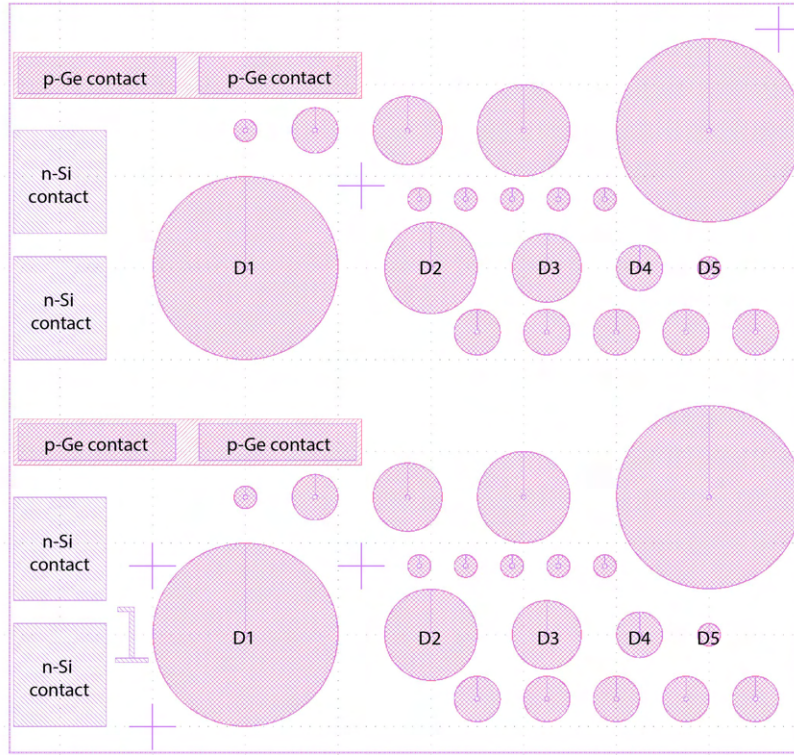


Figure 1-10: Example of diode mask design used in fabrication.

1.3.2 Measurement Setup

To characterize the diodes, two critical parameters are dark current density and quantum efficiency. A simple measurement setup was adjusted in order to provide reliable and consistent data for both. The Keithley 2450 source meter unit (SMU) applied a voltage and measured current measured, which was sufficient for dark I-V measurements. However, photocurrent measurements (from which quantum efficiency is calculated) preferentially have chopped light at each voltage. In other words, the light source would be continuously blocked and un-blocked relatively quickly (ex. 200 kHz) so that instantaneous photocurrent measurements could be extracted. A lock-in amplifier would then be used to extract the exact magnitude of the photocurrent, which in this case was the amplitude of a square wave.

Although setting up an optical chopper (Stanford SR540) and lock-in amplifier (Stanford SR810) was fairly straightforward, the photocurrent needed to be directed away from the Keithley SMU and towards the lock-in amplifier. [Figure 1-11](#) illustrates how this was accomplished with the use of a bias tee.

In the LTSpice schematic, a bias tee used a capacitor and inductor to separate DC and AC current, where AC current was essentially photocurrent and DC current was dark current. The photodiode was simulated as having 400 nA of dark current, and 100 nA of photoresponse, with the chopper set to a frequency of 200 kHz. The inductor was actually an 100 k Ω m resistor, which contained a parasitic inductance of 30 μ H. The lock-in amplifier was separated from the photodiode with a 6.8 μ F capacitor.

[Figure 1-12](#) shows the simulated photocurrent from an LTSpice simulation. The peak-to-peak of this square wave was the photocurrent; however, the lock-in amplifier did not output the peak-to-peak signal. Instead, it output the root mean square (rms) of the fundamental sine wave. In a square wave signal, this value is only 45% of the total signal. This percentage was also verified experimentally by comparing the lock-in signal to a manually collected instantaneous signal, from manually blocking/unblocking the light of a given diode with an IR card.

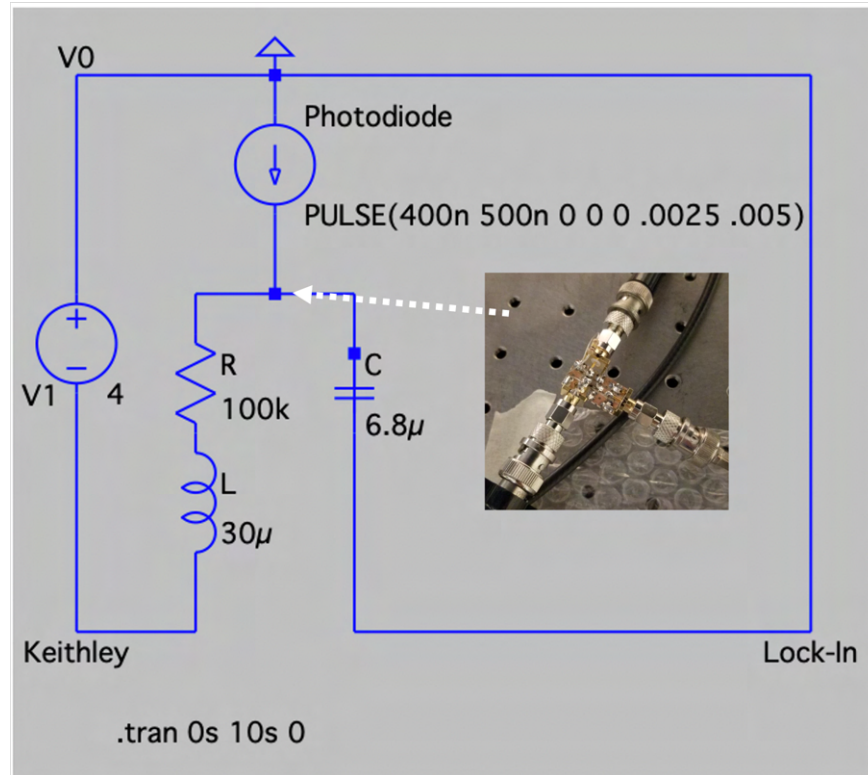


Figure 1-11: Equivalent circuit model for quantum efficiency and dark IV measurement setup, including the location and an image of the bias tee.

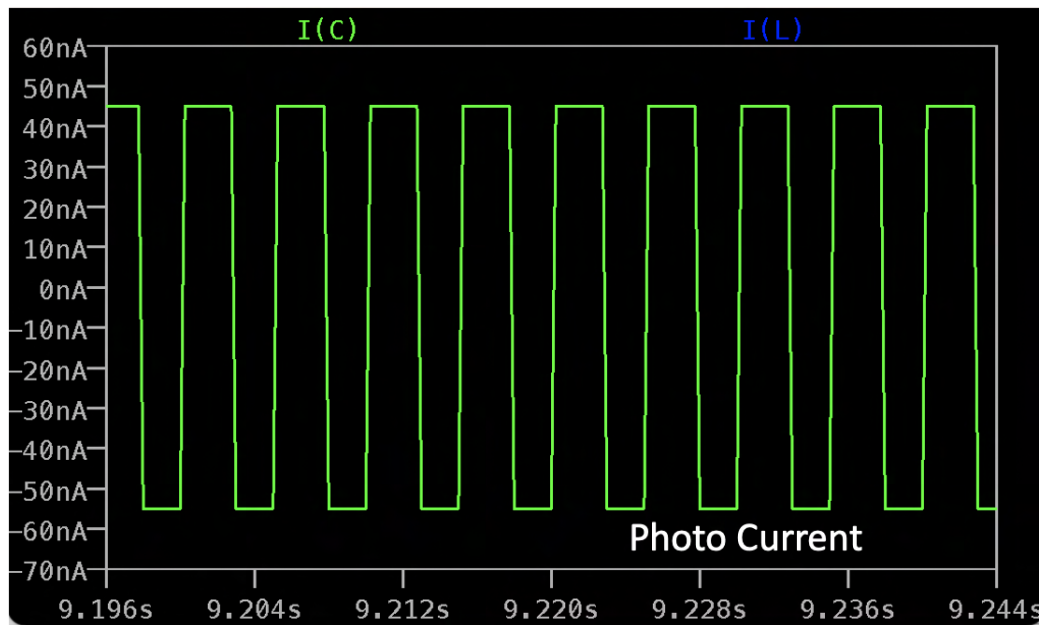


Figure 1-12: Simulated photocurrent from LTSpice, as seen by the lock-in amplifier.

For a 2 V peak-to-peak signal, a square wave may be broken down into the following sine components. Equation 1.7 and Equation 1.8 demonstrate how a lock-in amplifier with a 1 V peak-to-peak signal only outputs 0.45 V, or 45% of the total photoresponse.

$$S(t) = 1.273 \sin \omega t + 0.4244 \sin 3\omega t + 0.2546 \sin 5\omega t + \dots \quad (1.7)$$

$$\text{Signal} = \frac{1}{2} * \frac{1.273}{\sqrt{2}} = 0.45 \quad (1.8)$$

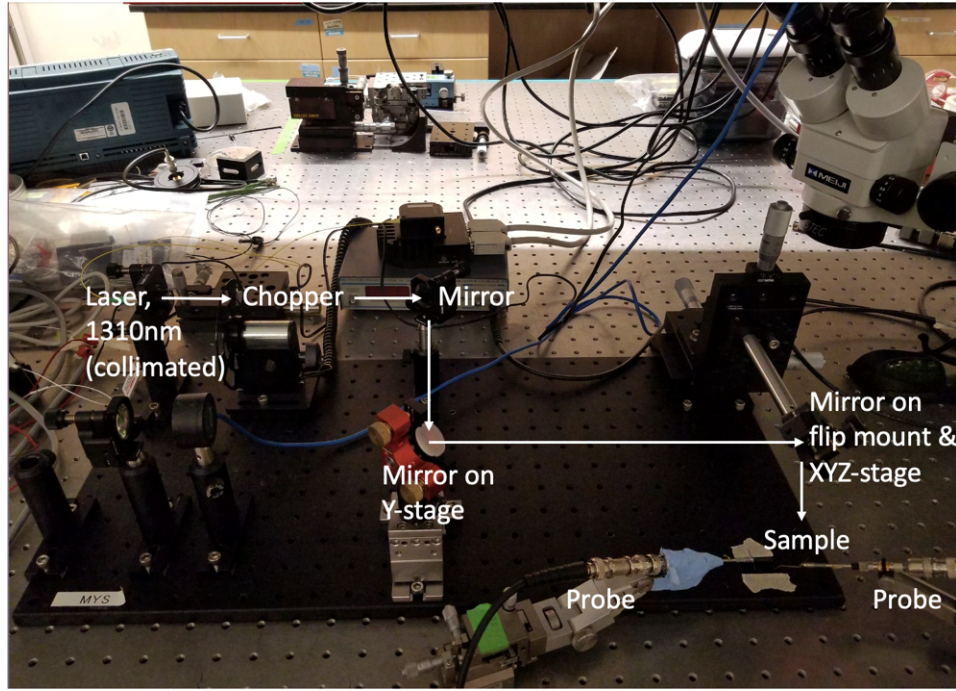


Figure 1-13: Photograph and schematic of the measurement setup for dark I-V and quantum efficiency.

With a reliable method for measuring photoresponse, the remainder of the optical bench setup is shown in Figure 1-13. As shown, a 1310 nm pigtailed laser diode with a collimated lens package passed through a chopper, impinged on a stationary mirror, and was then directed by two more mirrors (on translation stages) to the sample. The 1310 nm laser was swapped out for a visible laser during coarse alignment, and the translation stages allowed the 1310 nm beam to be aligned to a given sample by

maximizing the lock-in signal. Since the metal contacts were contacted via probes, the final mirror on the XYZ stage was placed on a flip mount to allow the microscope to move in and out of the setup as needed.

Quantum efficiency is a measure of the percentage of incoming light that generates photocurrent. Internal quantum efficiency (IQE) is based on the incoming light that actually gets absorbed, while external quantum efficiency (EQE) is based simply on the incident light. The calculations for IQE and EQE are given in [Equation 1.9](#) and [Equation 1.10](#), where α is the absorption coefficient of germanium, x is the thickness of the Ge epi layer, q is the charge on an electron, and R is the reflection at the diode surface.

$$\text{IQE} = \frac{\text{Photocurrent}/q}{\text{Incident Power} * (1 - R) * (1 - e^{-\alpha x}) / \text{Photon Energy}} \quad (1.9)$$

$$\text{EQE} = \frac{\text{Photocurrent}/q}{\text{Incident Power} / \text{Photon Energy}} \quad (1.10)$$

The measurement setup described provided an accurate photocurrent, but the incident power still needed to be measured accurately. Since the sample aperture size was not necessarily larger than the beam waist, the beam width required an accurate calibration. Using a Thor Labs M2MS beam profiler, which contained $5 \mu\text{m} \times 9 \text{mm}$ slits to perform a line integration that could be reconstructed in 2D/3D, the width of the Gaussian beam was found to be 2.28 mm. [Figure 1-14](#) below shows of an example of the profile in 2D, with its curve fit in the dashed line.

Finally, accounting for mirror reflections, the beam width, and a 1.16 mW intensity provided a final incident light power of $3.05 \mu\text{W}$ for the smallest diode, which had an exposed germanium area of $2.32 \times 10^{-4} \text{cm}^2$. The light intensity for a given area was then $0.013 \text{W}/\text{cm}^2$.

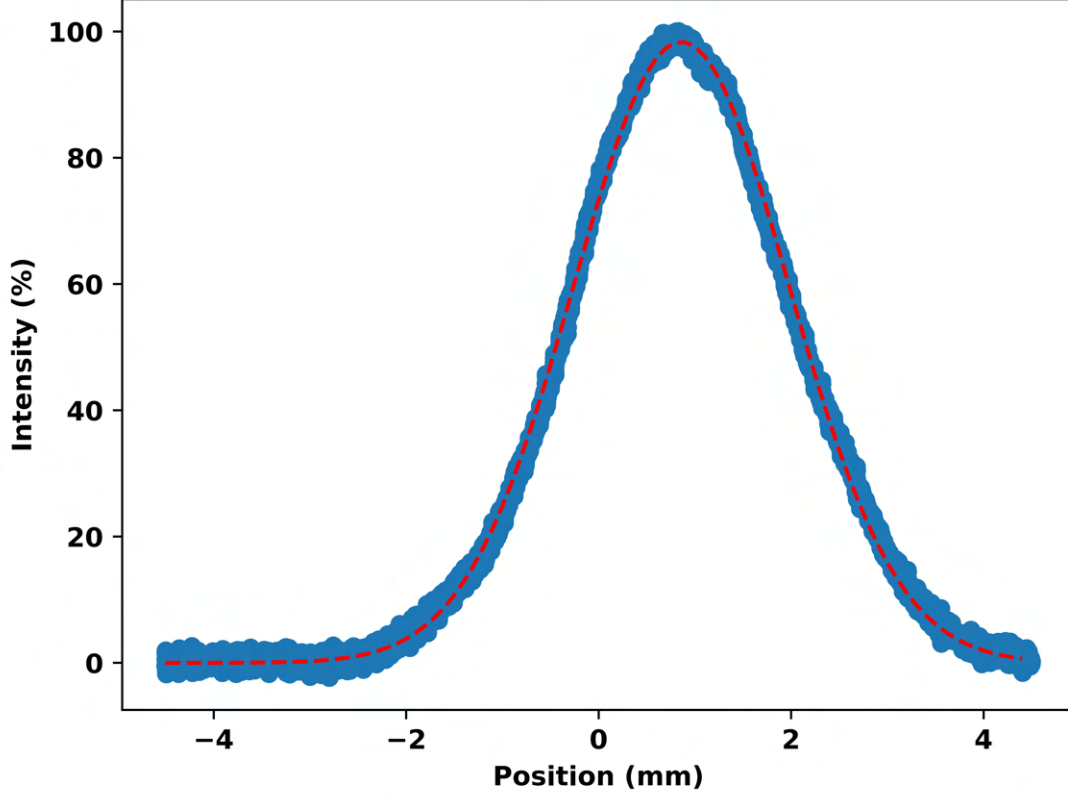


Figure 1-14: Thor Labs M2MS beam profile of the collimated 1310 nm laser.

1.3.3 I-V Characteristics

Dark current has contribution from both the diode area and periphery; however, we verified that the contribution was primarily areal and that the passivation was effective at reducing surface recombination velocity (SRV) at the dry-etched sidewalls. In [Figure 1-15](#), dark current is plotted against diode radius on a log-log plot. If the dark current were dependent primarily on the periphery, the dependence with diode radius would be r^1 , while a primary dependence on the area would yield a dependence of r^2 . Visually, the dark current appeared to increase by roughly two orders of magnitude for every magnitude of diode radius, and in fact the curve fits were close to r^2 dependence. The relations were $r^{1.7}$, $r^{1.8}$, and $r^{1.9}$ for Gen2 800 °C 2 min, Gen3 as grown, and Gen3 450 °C 4 hr, respectively.

A low temperature anneal at 450 °C reduced the total dark current by 210 nA for the smallest Gen3 diodes, translating to a reduction of $J_d = 107 \mu\text{A}/\text{cm}^2$ and

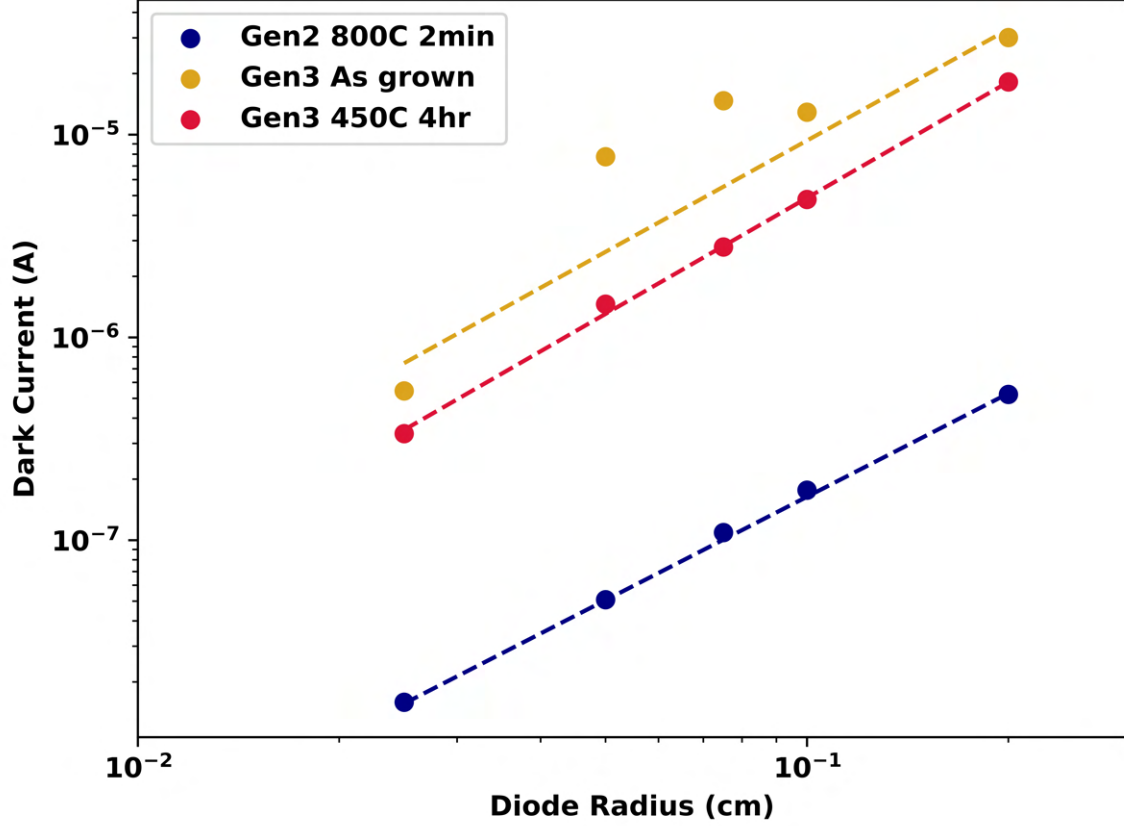


Figure 1-15: Effect of ALD passivation on dark current trend with diode radius. Dark current values were taken at $V_r = 1$ V for Gen2 800°C 2 min and 4 V for Gen3 diodes.

a relatively low dark current density of $J_d = 171 \mu\text{A}/\text{cm}^2$. The high temperature anneal for Gen2 resulted in the highest quality diode with a dark current of 16 nA and $J_d = 8 \mu\text{A}/\text{cm}^2$. Despite this achievement, even lower dark currents of $160 \pm 83 \text{ nA}/\text{cm}^2$ were previously obtained at the same reverse bias, using the same material and passivation method [11]. The reason for such discrepancy is not clear at this time, but the results reported in this thesis are based on processing that was done more than two years after the initial fabrication.

Although the diodes were passivated with 20 nm of Al_2O_3 , the radial dependency with dark current is not exactly r^2 , and the dry-etched periphery still contributes to the dark current density to some degree. Plotting J_d against perimeter/area allows one to extract the areal contribution, by looking at the y-intercept of the regression line as shown in Equation 1.11. Figure 1-16 shows that the previously confirmed

ALD passivation results in a stable J_d regardless of diode size, visually appearing as a nearly horizontal line with only a slight slope. For these diodes, J_A was found to be $4 \mu\text{A}/\text{cm}^2$ for Gen2 800 °C 2 min, and $149 \mu\text{A}/\text{cm}^2$ for Gen3 450 °C 4 hr. As expected, the areal contribution was only marginally lower than the total $J_d = 8 \mu\text{A}/\text{cm}^2$ and $J_d = 171 \mu\text{A}/\text{cm}^2$ for the respective diodes.

$$J_D = J_P(V) * \frac{\text{Perimeter}}{\text{Area}} + J_A(V) \quad (1.11)$$

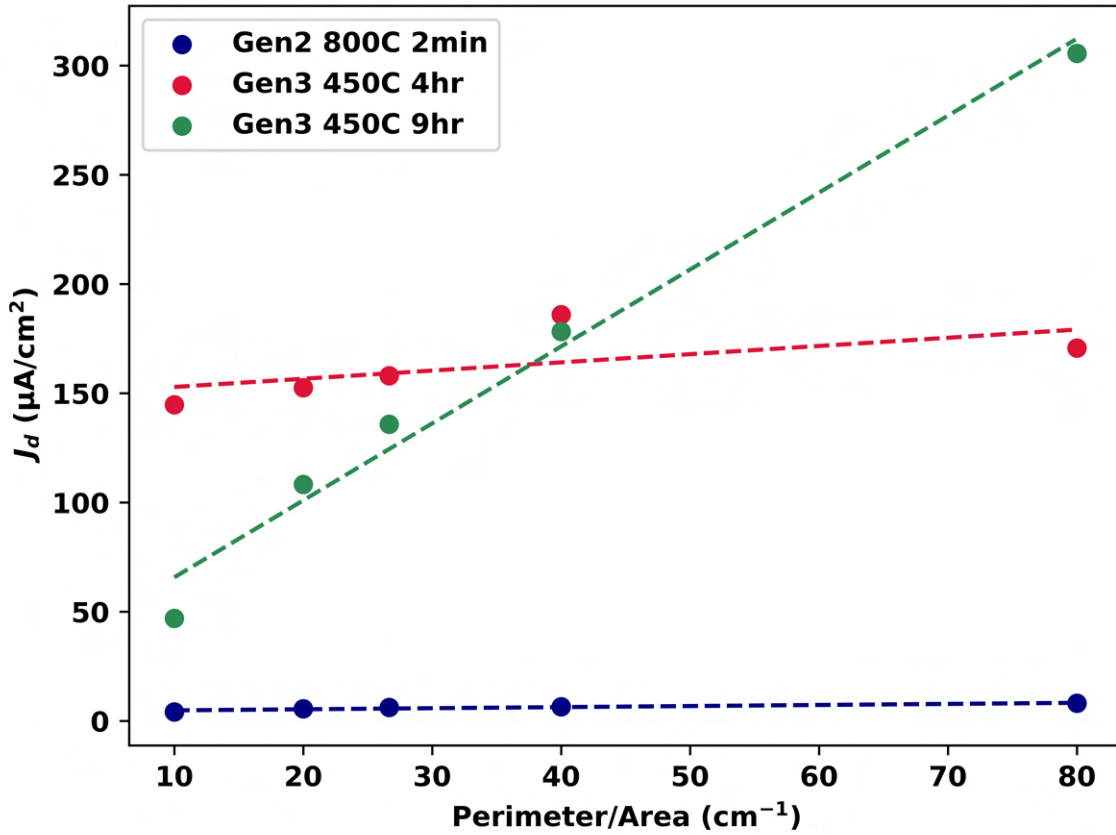


Figure 1-16: J_d plotted against perimeter/area to extract the contribution to dark current density from diode area. Dark current values were taken at $V_r = 1 \text{ V}$ for Gen2 800 °C 2 min and $V_r = 4 \text{ V}$ for Gen3 diodes.

An example of a diode suffering from high peripheral contribution was also plotted in Figure 1-16, as demonstrated by the steep slope for Gen3 450 °C 9 hr. An areal contribution of $31 \mu\text{A}/\text{cm}^2$ at $V_r = 4 \text{ V}$ was extracted, which is a shocking factor of 10x smaller than the total $J_d = 305 \mu\text{A}/\text{cm}^2$ for the smallest diodes measured.

Considering more closely the total areal contribution to leakage current, the diode top surface containing 2 nm Si cap was evaluated to ensure it was not having any negative impact on performance. A set of Gen2 and Gen3 diodes were processed in the same fabrication lot, but one set of diodes had the Si cap removed at the beginning via 2.5% TMAH dip for 1 min, followed quickly by the ALD deposition, while the second set underwent the regular process flow in [Table 1.2](#). A comparison of dark I-V characteristics in [Figure 1-17](#) for the Gen2 and Gen3 diodes demonstrated that the Si cap has little impact on leakage current. These results helped validate the process flow, as every diode measured in this work contained an Si cap with the exception of Gen2 800 °C 2 min. The Si cap layer was also kept intentionally to protect the somewhat reactive Ge surface during fabrication, for example from etching by DI water. Literature findings suggested that Si epitaxy layers on Ge could undergo strain relaxation as early as 0.8 nm and 1.4 nm via 90 ° partial and 60 ° perfect dislocations respectively [12]. The Si caps for these wafers were not grown epitaxially, but were likely not entirely amorphous. Fortunately, even past the turn-on voltage of $V_r = 4$ V, any Gen2 or Gen3 diode with an Si cap had either the same or lower J_d than its counterpart without the Si cap. The Gen3 sample with Si cap appeared to have a higher series resistance than the rest, likely from a fabrication abnormality, and this would have effectively elongated the turn-on voltage region as shown in the figure.

For pn diodes, series resistance should be minimized and shunt resistance maximized for best performance. These characteristics were confirmed by analyzing the I-V curves at specific voltages. In forward bias after the initial increase in current, the series resistance is equivalent to the inverse slope. The shunt resistance is equivalent to the inverse slope around zero volts, with the idea that any dark current close to zero voltage comes not from diode behavior but leakage current on the periphery, a sign of low shunt resistance. As shown in [Figure 1-18](#), the measured diodes exhibited a high shunt resistance of 13.7 M Ω and a low series resistance of 0.61 m Ω .

The diode capacitance also scales with area as expected in [Figure 1-19](#), and is nearly identical for Gen2 and Gen3 for the annealing conditions as shown in the figure below. The estimated depletion width at zero bias is 0.81 μ m, and the capacitance

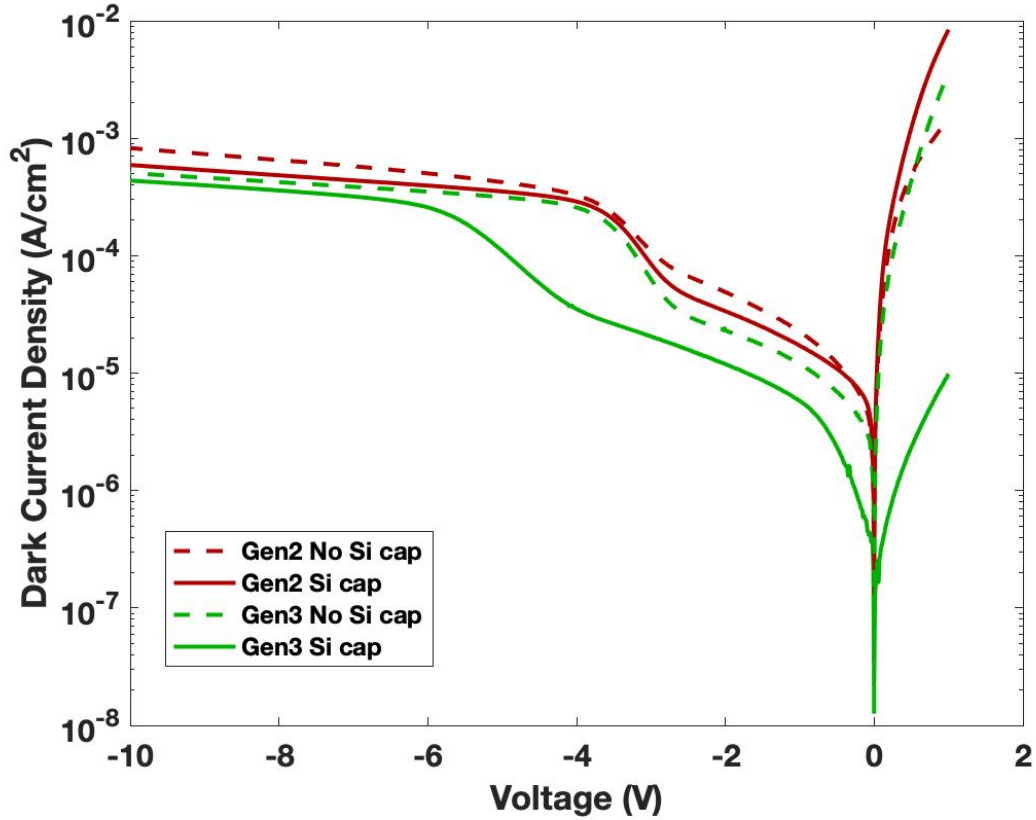


Figure 1-17: Effect of Si cap on J_d for Gen2 and Gen3 diodes.

for the 250 μm diodes was 25 pF.

A set of diodes that experienced annealing conditions ranging from as-grown up to 550 °C for multiple hours were directly compared in Figure 1-20. After overcoming the energy barrier created by the doping profile at -4 V, the dark current density quickly saturated in most of the diodes to a value as low as 275 $\mu\text{A}/\text{cm}^2$. With the exception of Gen3 500 °C 3 hr, the Gen2 diodes demonstrated lower dark current densities than the Gen3 diodes. The dark I-V curves mostly followed the SRH model for a pn diode after overcoming the energy barrier, but the diodes that were annealed at 550 °C did not saturate, but instead continued to increase with reverse bias by as much as one order of magnitude in J_d for Gen3 550 °C 4.25 hr.

Although the increase in J_d with reverse bias superficially appears to be related to annealing conditions, at least one reason for this effect may simply be a slight difference in fabrication between two lots of samples. The as-grown and 500 °C diodes

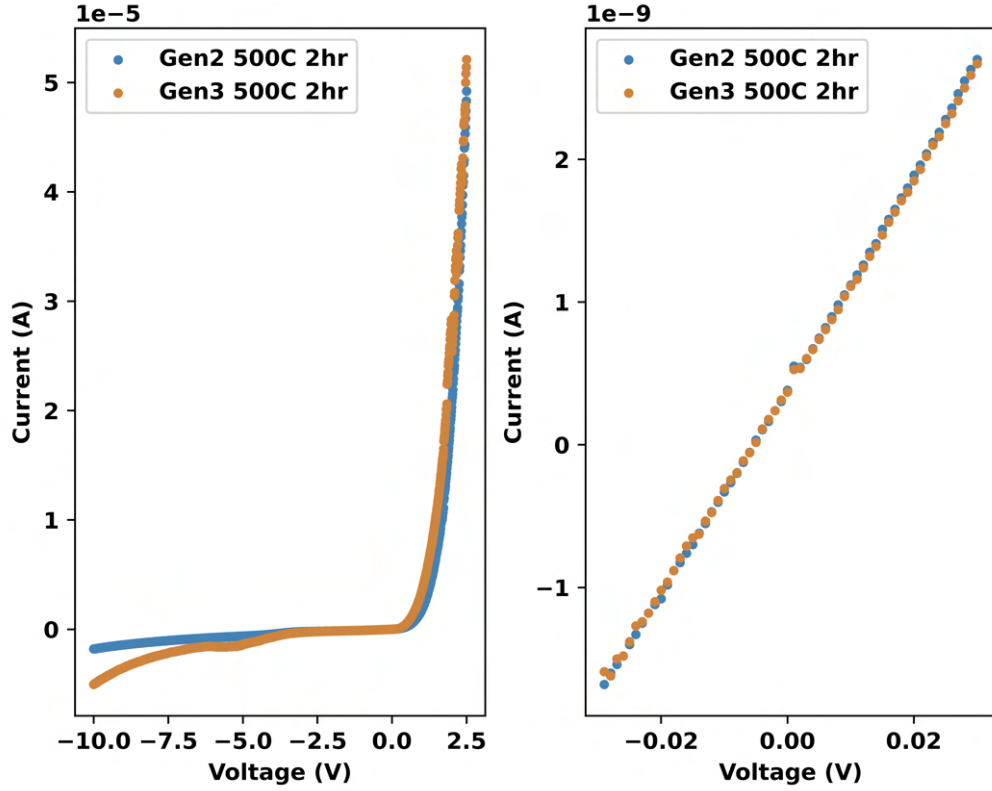


Figure 1-18: Dark I-V measurements used to extract series and shunt resistance for Gen2 and Gen3 devices annealed at 500 °C for 2 hr.

were processed in a separate lot from the 550 °C diodes, and the latter were fabricated alongside the Gen3 450 °C 9 hr samples that showed relatively low areal contribution and high peripheral leakage in [Figure 1-16](#). Any peripheral leakage paths increase the linear dependency of V_r and J_d , which prevents the dark current from saturating as expected in the SRH model. In contrast, the as-grown and 500 °C diodes were fabricated in the same lot as the Gen3 450 °C 4 hr samples, which showed dominant areal contribution and negligible peripheral leakage in the same [Figure 1-16](#).

The high peripheral leakage in the second lot could either be caused by contamination of the ALD chamber, or difference in etch equipment. The latter would be the more suspect root cause, since the first set of samples were dry-etched in the RIE- Cl_2 -Samco, which only accepts standard electronic materials, and the second set of samples were dry-etched in the RIE-Mixed-Samco, which accepts all R&D materials. A contaminated dry etch would inhibit correct passivation during ALD and result in

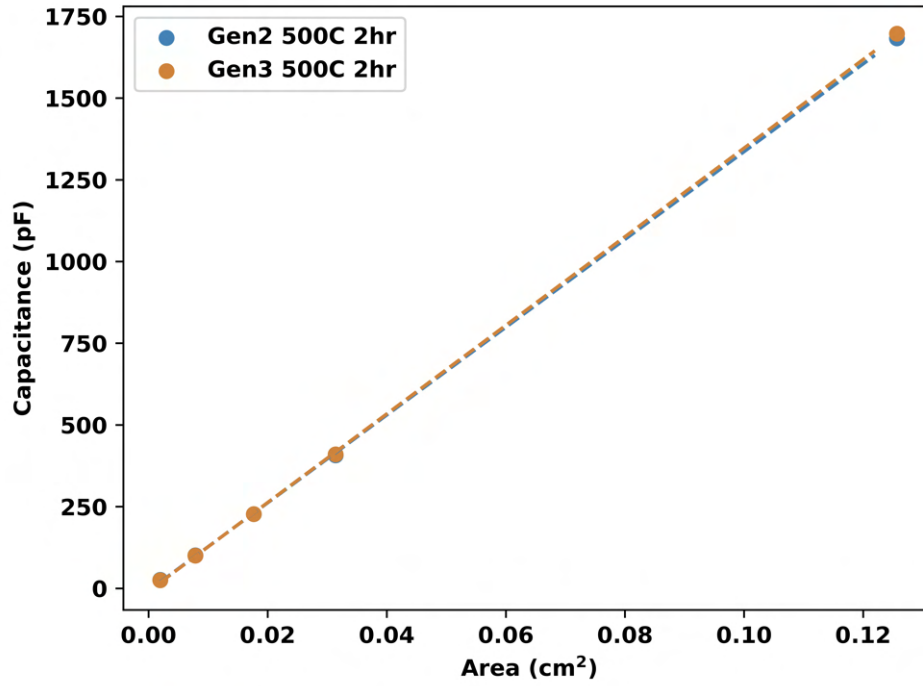


Figure 1-19: Capacitance measurements taken at zero bias, for Gen2 and Gen3 devices annealed at 500 °C for 2 hr.

high peripheral leakage. The only reason for switching etch equipment was to process more samples simultaneously - the RIE-Cl₂-Samco tended to form "black grass" if more than one sample was loaded in the etch chamber. In the future, more consistent verification of areal versus peripheral contribution in a given sample lot is important for verifying device quality. The high shunt resistances in [Figure 1-18](#) were indicative of low peripheral leakage, but the samples were from a different fabrication lot entirely, and the relationship between shunt resistance and peripheral contribution has not been directly correlated yet in this work.

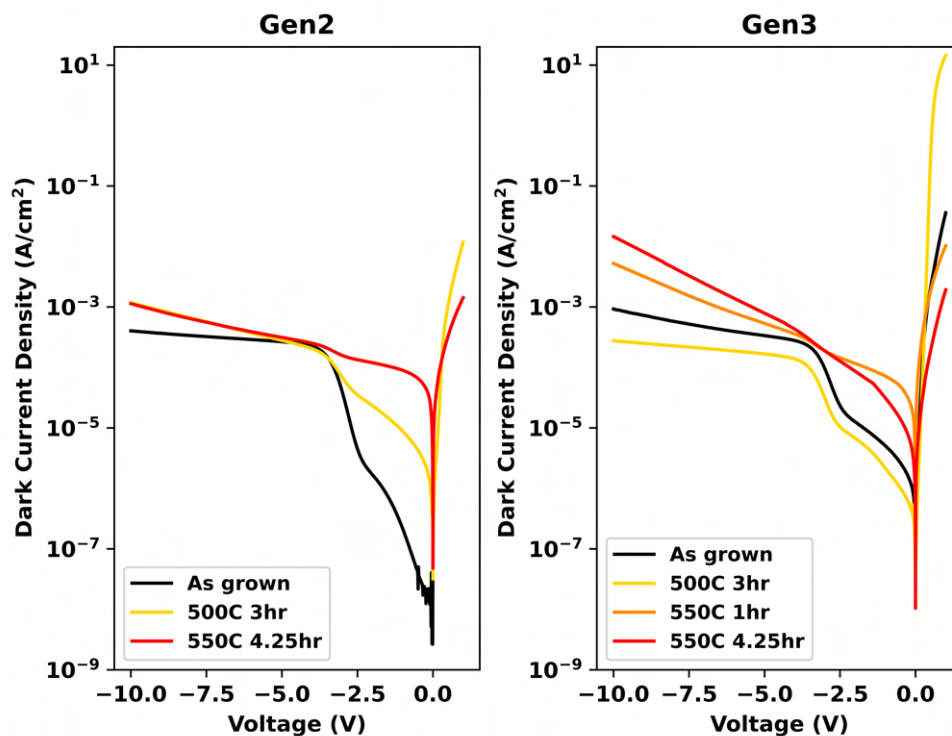


Figure 1-20: Dark I-V scans at various annealing conditions for Gen2 and Gen3 devices.

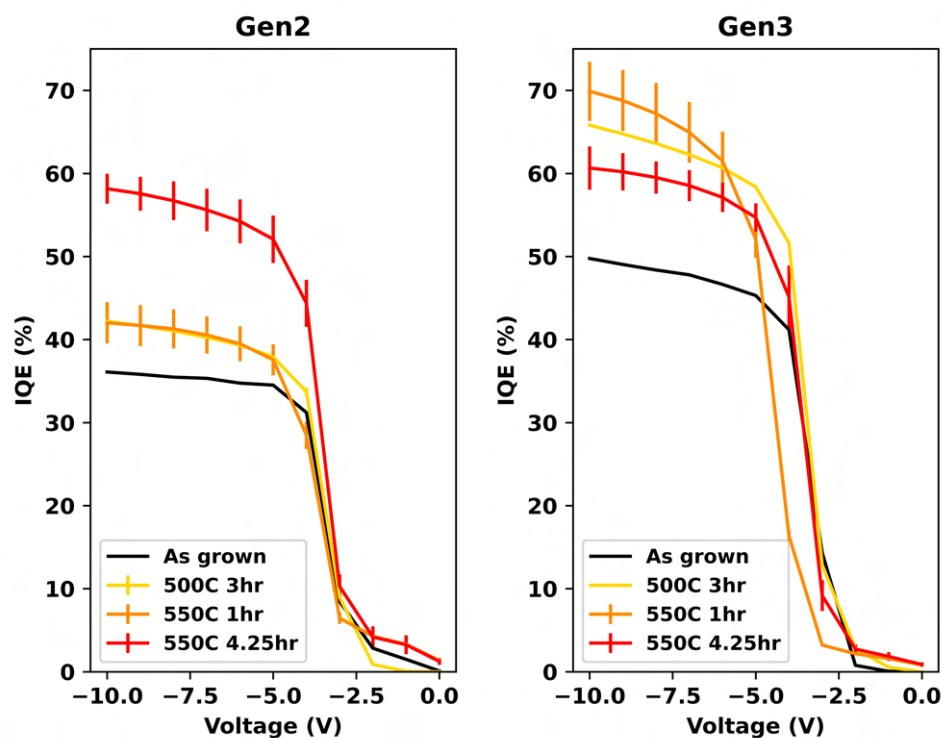


Figure 1-21: IQE at various annealing conditions for Gen2 and Gen3 devices.

The effects of annealing on IQE are shown in [Figure 1-21](#). The Gen3 diodes had higher IQE than the Gen2 diodes, which was unsurprising given the presence of the built-in E field. Regardless, the IQE of the as-grown diodes was already surprisingly high, with Gen2 and Gen3 starting at an IQE of 36% and 50%, respectively. Higher annealing conditions were required for Gen2 to attain nearly as good performance as Gen3, but ultimately the IQE reached as high as 58% for Gen2 at 550 °C for 4.25hr and 70% for Gen3 at 550 °C for 1 hr. The IQE improved consistently with increased annealing temperatures and times, with the 550 °C for Gen3 serving as a singular exception. The demonstrated improvement in IQE with annealing was likely a result of point defect annihilation, which would have an associated activation energy [11], while the unexpected reduction in IQE at the highest annealing time and temperature is perhaps a different process. Future research can look more closely into the possible defect mechanisms involved, and provide a model to support these experimental results.

The dark current densities for germanium on silicon diodes in this work were some of the lowest reported in literature [13] [14] [15] [16], and these low dark current values were coupled with high quantum efficiency. In particular, as-grown conditions for Gen3 resulted in a high IQE of 45% at $V_r = 5\text{ V}$, and $J_d = 340\text{ }\mu\text{A}/\text{cm}^2$. Annealing at 500 °C for 3 hr improved the IQE to 58% at the same reverse bias, and lowered the dark current density to $J_d = 170\text{ }\mu\text{A}/\text{cm}^2$. Low temperature germanium-on-silicon grown photodetectors, with or without post-anneals have proven to be effective and have direct potential for incorporation into back-end-of-line processing. A direct comparison of responsivity and J_d values from this work and literature is detailed in the Conclusions.

1.4 Anti-Reflective Coating and RFPCD

Although it is informative to evaluate IQE, which is given relative to absorbed power as opposed to incoming power, it is advantageous from an engineering perspective for the device to perform efficiently in both cases. The inclusion of an anti-reflective (AR) coating in the design minimizes the surface reflection of incoming light and increases the real-life performance of the photodiode.

Since 1310 nm light is absorbed in the Ge epilayer, the design can essentially be treated as an AR coating on Ge substrate. A three-layer design is shown in [Figure 1-22](#), and the theoretical performance with and without the 2 nm Si cap layer is shown in [Figure 1-23](#). The design was optimized for the Si cap layer, but the reflectance at 1310 nm is essentially the same without it, approximately 0.4%.

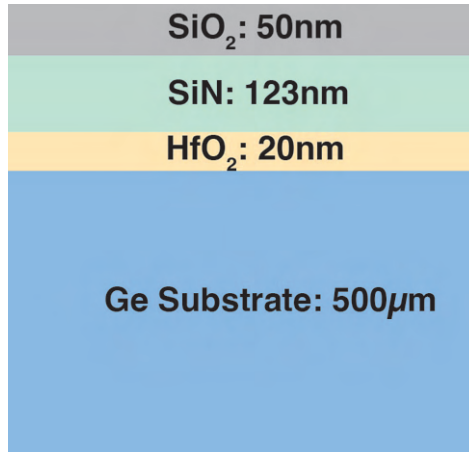


Figure 1-22: Antireflective (AR) coating design for Ge substrate at 1310 nm.

To demonstrate proof-of-concept, two Ge substrates each 500 μm thick were immersed in BOE 1:7 for 12 minutes to remove the 20 nm Al_2O_3 layer that had been previously coated. One substrate was then set aside to serve as a control sample, while the other was rinsed with water, air-dried and promptly loaded into the ALD-Ozone chamber. The HfO_2 coating was deposited at 200 $^\circ\text{C}$, followed by low-stress PECVD depositions of SiN and SiO_2 . The AR coating dimensions were designed to be easily fabricated in a reasonable time frame via ALD and PECVD processes.

[Figure 1-24](#) shows the FTIR reflectance of the AR coating on Ge substrate, which reached 4.7% at 1310 nm. Part of the reason for the higher than expected reflectance

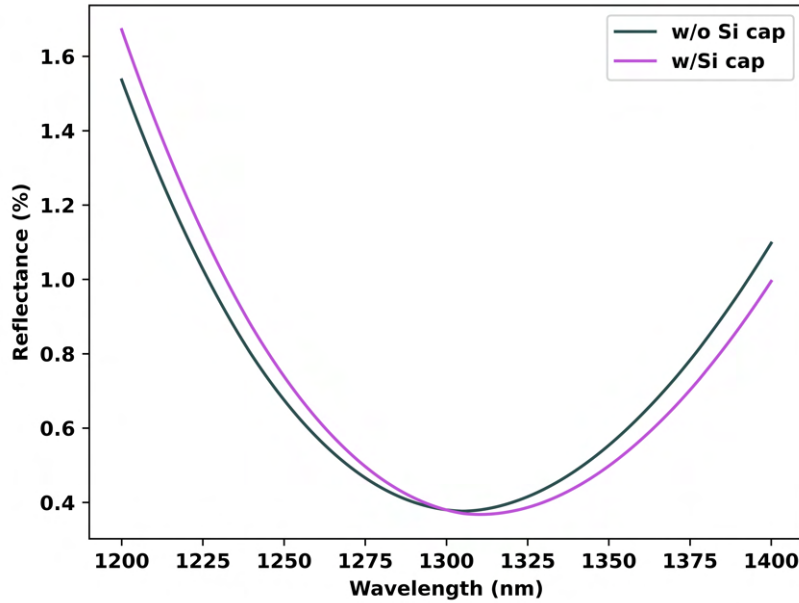


Figure 1-23: Reflectance of theoretical AR coating design with wavelength.

is due to the layer thicknesses being slightly off from theoretical value, with the greatest discrepancy in the SiN layer. However, this alone was not able to account for the high reflectance reading. Since FTIR is not strictly performed at near-normal incidence, matched theoretical designs were plotted on the same graph at 30 ° angle-of-incidence (AOI) and 0 ° AOI. These reflectance curves were designed to closely match the measured spectral results. Although the minima appear at the same wavelength, the absolute reflectance values from the FTIR are consistently higher. Possibly there was an issue with a baseline instrument scan, but still a relatively low reflectance of 4.7% is demonstrated at the target wavelength, with an even lower reflectance of 1.0% from the matched theoretical design and 0.4% from the original theoretical design with perfect execution.

Ellipsometry were also performed to provide additional verification of each layer thickness. As shown in [Table 1.3](#), the measured values from ellipsometry were close to the estimated values from the matched design, with at most a 6 nm discrepancy for the SiN layer. The Cauchy and Urbach fit provided the same result as the Sellmeier extreme angle fit, with R^2 values of 98.6% and 97.7%, respectively. Ellipsometry and FTIR scans were performed by Cosmin Popescu.

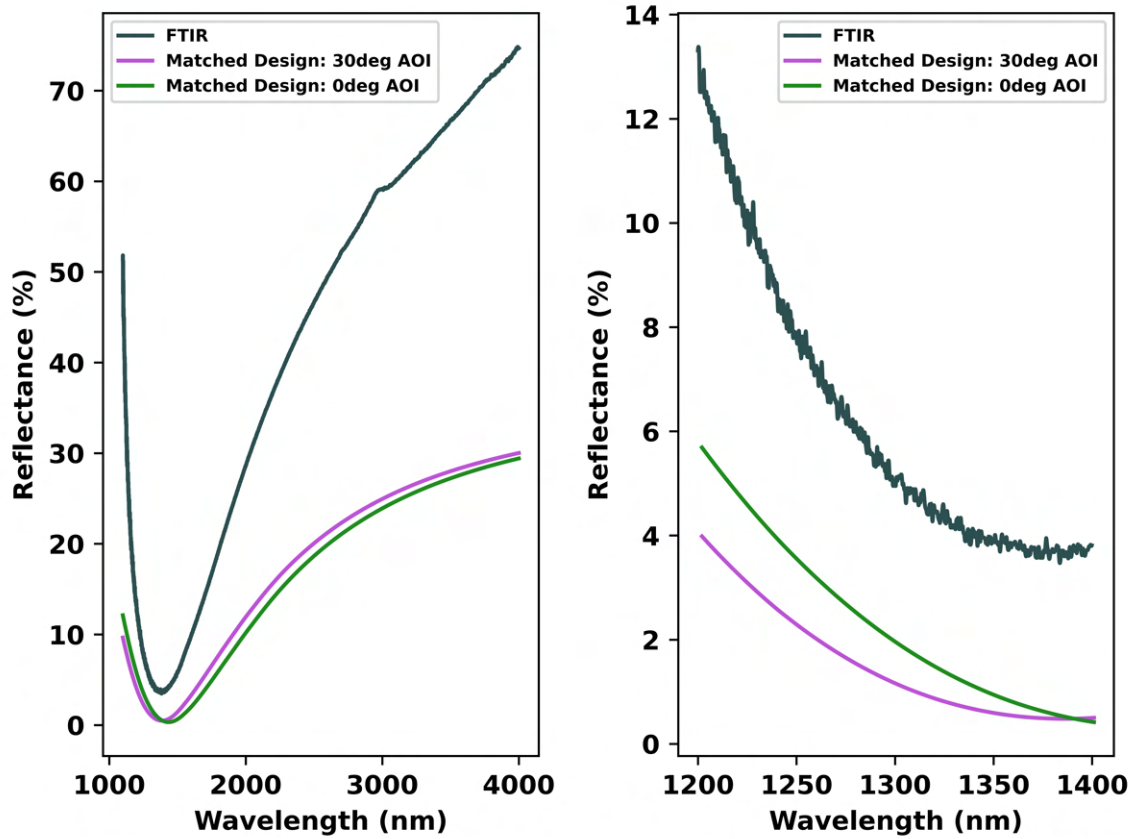


Figure 1-24: Fourier-transform infrared spectroscopy (FTIR) measurements of deposited AR coating on Ge substrate.

Table 1.3: Target and Actual AR Coating Layer Stack.

Layer Thickness (nm)	Target	Actual (from ellipsometry)	Actual (from matched design)
SiO ₂	50	49	50
SiN	123	146	138
HfO ₂	20	25	22

In addition to confirming that surface reflection had been effectively minimized at 1310 nm, the impact of the AR coating process on surface quality needed to be verified. Radio frequency photoconductive decay (RFPCD) measurements were taken on both the control sample and the AR coated sample to evaluate the effects of the deposition processes on carrier lifetime. RFPCD is setup with an RF coil to inductively measure the conductance of the wafer, while the substrate is pulse-illuminated with visible light from the topside. The light illumination generates excess free carriers, which have a

measurable exponential decay allowing for accurate measurements of carrier lifetime. Figure 1-25 showed how a light that is turned off contains a turn-off transient, with rate of exponential decrease determined by the carrier lifetime of the material. This textbook example is for n-substrate, thus the excess minority carrier concentration is given as $p'(t)$ [7]. For Ge substrate, blue and red LEDs (850 nm) were used for carrier generation, with penetration depths of 20 nm and 200 nm, respectively. With a 1310 nm AR coating, only 10-15% of blue light and 30-35% of 850 nm light was reflected from the surface.

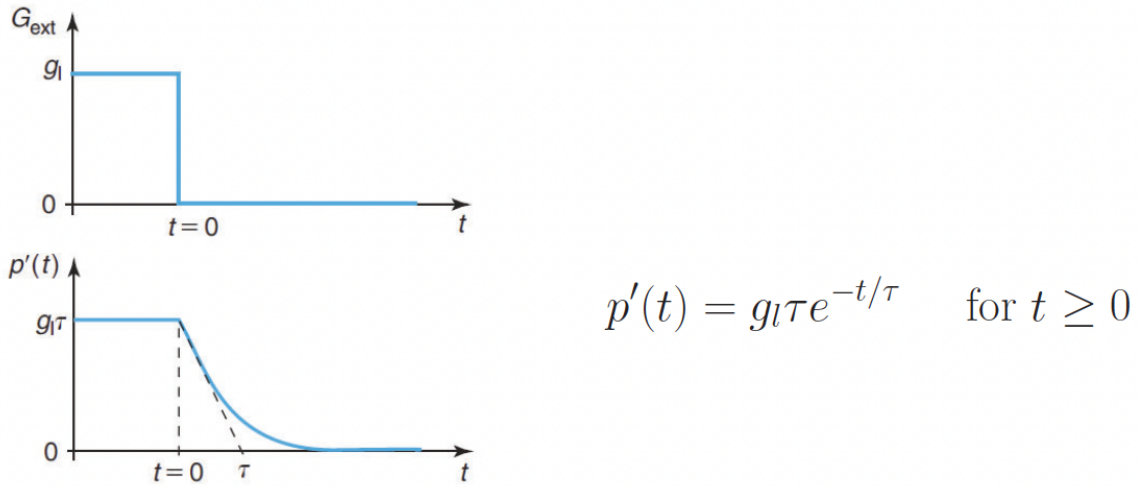


Figure 1-25: Excess carrier generation with a pulse of light, and exponential relation with carrier lifetime [7].

In Table 1.4, the AR coating had clear impact on carrier lifetime at the surface. Post-fabrication, the unprocessed Ge substrate and the coated Ge substrate were measured at approximately the same time, but the control sample measured $554 \mu\text{s}$, more than 10x higher than the coated Ge substrate. The frontside and backside measurements on the Ge substrate were nearly identical, which was expected given that the diffusion length of carriers in bulk Ge substrate is significantly greater than the wafer thickness. The LED color did not have significant impact on the carrier lifetime measurements. However, the carrier lifetime for the control sample dropped by almost $100 \mu\text{s}$ just 2 hr after fabrication, in contrast to the much more stable, albeit lower carrier lifetime of $45 \mu\text{s}$ in the AR-coated sample.

Table 1.4: Radio frequency photoconductive decay (RFPCD) for Ge Substrate w/ and w/o AR Coating.

Carrier Lifetime (μs) λ	Post-Fabrication Blue Light	Post-Fabrication + 2 hr Blue Light	Post-Fabrication + 2 hr 850nm
Ge Substrate w/o AR	554 ± 19	462 ± 9	481 ± 23
Ge Substrate w/AR, frontside	47 ± 4	45 ± 1	46 ± 2
Ge substrate w/AR, backside	45 ± 2	46 ± 2	45 ± 1

The surface recombination velocity of the AR-coated surface may be extracted from the effective carrier lifetimes that have been measured as shown in [Equation 1.12](#) [17]. Although this should typically be applied only in scenarios where both surfaces are created equal, it serves as a good estimate of surface quality. The SRV of 5.55×10^2 cm/s for the AR-coated substrate is low enough to yield good device performance.

$$\tau_{\text{eff}} = \frac{d}{2S} \quad (1.12)$$

1.5 Conclusions

Near-infrared detection is widely used in optical communication, imaging, and spectroscopy [18]. The current limitations of optoelectronic integration with Si CMOS circuits reduce design flexibility and increase device size. The metal interconnect layers, or the back-end-of-line (BEOL) serve as the ideal placement for these photodetectors; however, a low thermal budget of 450 °C is required [19]. In this work, we demonstrated that BEOL-compatible, vertical incidence Ge-on-Si photodiodes are competitive with high temperature-grown or annealed devices, and we explored low temperature anneals up to 550 °C. We also demonstrated the efficacy of the ALD passivation method to obtain a dark current dependence with area instead of periphery, and tested an anti-reflective coating that would increase the EQE and responsivity by 62%.

The best performing diodes, Gen3 contained a built-in E field within the Ge layer via graded doping, promoting a drift current towards the junction and reducing surface recombination. Although certain annealing conditions improved IQE, as-grown (unannealed) conditions also resulted in a high IQE of 45% at a reverse bias of $V_r = 5$ V, saturating to $\text{IQE} = 50\%$ at $V_r = 10$ V. The 500 °C and 550 °C anneals improved the IQE by 7-20% in the voltage range $V_r = 5$ -10 V. The delayed onset in IQE may be attributed to the doping transitions from epitaxial growth, including the transition region shown in the cross-section, and can be tuned to have an earlier turn-on voltage. The dark current densities for the as-grown and 500 °C 3 hr annealed diodes were relatively low: $J_d = 340 \mu\text{A}/\text{cm}^2$ and $170 \mu\text{A}/\text{cm}^2$ at $V_r = 5$ V.

Responsivities for as-grown and 500 °C 3 hr annealed diodes were 0.11 A/W and 0.15 A/W at $V_r = 5$ V. Incorporating the anti-reflective coating would increase such values to 0.19 A/W and 0.24 A/W, and complete absorption in the 500 nm Ge epi would further increase to 0.48 A/W and 0.62 A/W, respectively. A comparison with other Ge-on-Si photodiodes from literature demonstrates competitive responsivity and dark current density, despite the low temperature growth and annealing conditions. In Table 1.5, the lowest processing temperature for any Ge-on-Si photodiode

in literature was 750 °C, with Osmond reporting the best result of $J_d = 60 \mu\text{A}/\text{cm}^2$ and 0.4 A/W responsivity at $V_r = 3 \text{ V}$, and Samavedam reporting a similar result to our work of $J_d = 150 \mu\text{A}/\text{cm}^2$ and 0.13 A/W.

Table 1.5: Comparison of near-infrared Ge-on-Si photodiodes from literature: Performance and process parameters.

	This work (2022)	Osmond (2009) [13]	Samavedam (1998) [14]	Lin (2021) [15]	Piels (2012) [16]
J_d ($\mu\text{A}/\text{cm}^2$)	170	60**	150	780	$3.5 \cdot 10^4$
Responsivity (A/W)	0.15	0.4	0.13***	0.29	0.29
V_r (V)	5	3	1	1	1
Max Process Temperature (°C)	500	780	750	750*	< 800
Ge (μm)	0.5	1.0	0.7	0.7	0.8
AR Coating	N	Y	N	N	Y
Wavelength (nm)	1310	1550	1300	1550	1310

Comments on [Table 1.5](#):

*Lin (2021): Ge epi annealed at 750 °C, but process was designed for low-temperature bonding and layer transfer at 300 °C (BEOL-compatible).

**Osmond (2009): Dark current density at $V_r = 3 \text{ V}$ estimated from Fig. 1 in article.

***Samavedam (1998): Responsivity measured in photovoltaic mode.

In conclusion, Ge-on-Si p-i-n photodetectors fabricated at BEOL-compatible temperatures (< 450 °C) have demonstrated comparable responsivity and dark current density to devices processed at high temperatures, and relatively low temperature anneals (500 °C) increased performance. As-grown diodes showed a responsivity of 0.11 A/W and $J_d = 340 \mu\text{A}/\text{cm}^2$, and annealing conditions of 500 °C 3 hr improved such performance to 0.15 A/W and $J_d = 170 \mu\text{A}/\text{cm}^2$ [20].

Chapter 2

MWIR waveguide integrated

InAs_xSb_{1-x}

2.1 Introduction

Mid-wave infrared (MWIR) photodetection has been successfully implemented for decades using the II-VI material set, Hg_{1-x}Cd_xTe. With a focus on military applications, extensive research was performed to maximize not just device performance, but also operating temperatures, manufacturability and cost. Such work has pushed HgCdTe to nearly reach its theoretical performance limit, while also highlighting its inherent shortcomings for commercialization. An upcoming material set, InAs_xSb_{1-x} has the potential to overcome such barriers while promising comparable performance [21].

The InAs_xSb_{1-x} alloy is a direct bandgap material with good tunability, high electron mobility, high optical absorption and low thermal generation [21]. Early growth methods such as liquid phase epitaxy [22] have since progressed to more advanced epitaxial methods including metal organic chemical vapor deposition (MOCVD) and molecular beam epitaxy (MBE). To realize its commercial potential in applications such as gas-mapping LIDAR and chem-bio sensing, InAs_xSb_{1-x} should be integrated heteroepitaxially with existing CMOS platforms; however, a significant lattice mismatch needs to be overcome. While Ge has a lattice constant of 5.658 Å, InAs and

InSb have lattice constants of 6.058 and 6.479 Å, respectively [21]. In addition, the group IV/III-V interface is also challenging to fabricate.

To resolve both issues at the interface, a GaAs buffer epilayer, with lattice constant of 5.653 Å may be introduced [21]. The close lattice match of Ge and GaAs essentially introduces a new III-V growth surface. Since $\text{In}_{0.91}\text{AsSb}_{0.09}$ has been successfully grown on GaSb substrate [23], there is also an option of incorporating GaSb as a second buffer epilayer if needed. Incorporating two III-V buffer layers on Ge for $\text{InAs}_x\text{Sb}_{1-x}$ growth essentially moves one step beyond a fabrication method already performed for $\text{InAs}_x\text{Sb}_{1-x}$ photodetectors on GaAs substrates via MOCVD, which incorporated both a GaAs buffer layer and a GaSb nucleation layer [24]. Another possible method for overcoming the lattice mismatch involves tensile straining the Ge waveguide with an SiN stressor layer, bringing the lattice constant closer to $\text{InAs}_x\text{Sb}_{1-x}$ [25].

2.2 Photodetector Simulations

For a waveguide-integrated photodetector design, optical constants needed to be defined for $\text{InAs}_x\text{Sb}_{1-x}$. Designing for the 3-5 μm range necessitates a bandgap of less than 0.25 eV, which requires careful tuning of the alloy composition. Experimental results from liquid phase epitaxy of $\text{InAs}_x\text{Sb}_{1-x}$ showed that the arsenide/antimony ratio must be at least 0.85/0.15, thus dictating the alloy composition $\text{InAs}_{0.85}\text{Sb}_{0.15}$. This composition has an optical absorption coefficient, $\alpha = 10^3 \text{ cm}^{-1}$ at 5 μm , promising good absorption in the desired photodetection range. Higher arsenide content results in even higher absorption at a given wavelength [22].

The refractive index was estimated based on a numerical Kramers-Kronig analysis of narrow bandgap III-V semiconductors. It was found that the presence of doping, or free carriers had a much greater impact on refractive index for n-type materials. For a 4.6 μm wavelength, the refractive index for intrinsic or 10^{18} p-doped $\text{InAs}_{0.85}\text{Sb}_{0.15}$ were both approximately 3.85, while 10^{18} n-doped material had a lower refractive index of 3.81. The overall range for refractive index at this wavelength ranged from

3.80 to 4.02, based on the alloy composition [26], and for simplicity 3.88 was selected as the refractive index for all subsequent photodetector simulations regardless of doping type or concentration.

For the photodetector design, two options were considered: homojunction and heterojunction $\text{InAs}_{0.85}\text{Sb}_{0.15}$ on Ge. The heterojunction design had higher potential for minimizing dark current, but an electronic band structure that was more difficult to successfully realize. As shown in Figure 2-2, a semimetal junction is formed at the GaSb/ $\text{InAs}_{0.85}\text{Sb}_{0.15}$ interface, which plays no adverse role in a homojunction design but serves as a barrier to carrier collection in a heterojunction design. Thus, the homojunction design was selected for subsequent photodetector simulations performed in Lumerical.

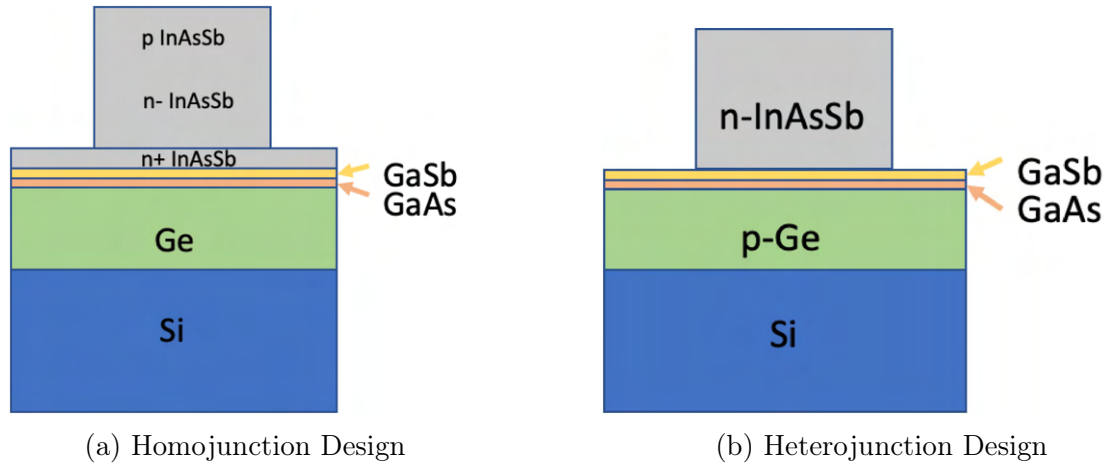


Figure 2-1: MWIR photodetector designs for waveguide integrated $\text{InAs}_{0.85}\text{Sb}_{0.15}$.

The GaSb and GaAs layers were included as buffer layers to allow for a successful epitaxy growth. Simulations demonstrated that the presence of any buffer layers reduced photodetector performance, and the thicknesses should be minimized. Each GaSb and GaAs buffer layer was set to 100 nm for the following simulations.

The Ge waveguide thickness was set to $1\ \mu\text{m}$, as the $0.5\ \mu\text{m}$ option from the foundry did not confine the mode well. Figure 2-3a includes example dimensions, and the bird's eye view in Figure 2-3b shows a taper from the input Ge waveguide to the photodetector region. Gold top contacts were used for carrier collection.

As a starting point, FDE (finite difference eigenmode) simulations were used to

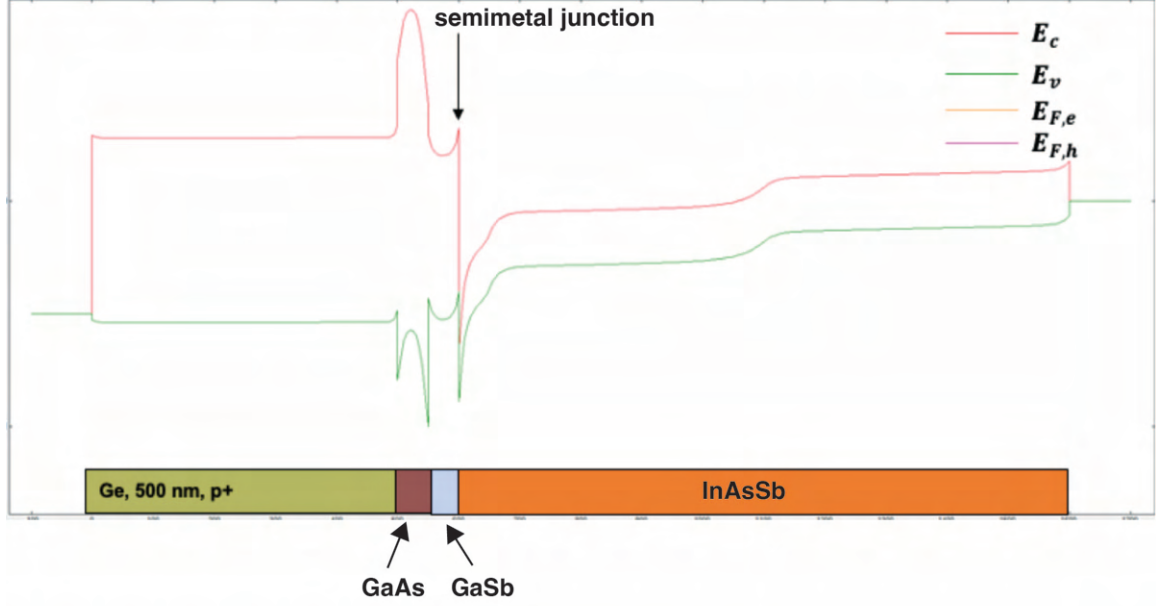
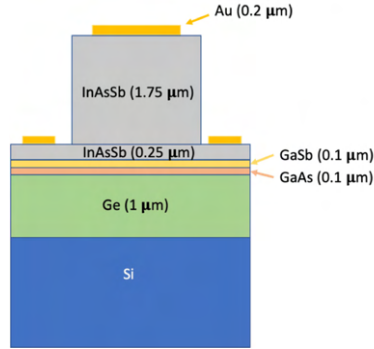
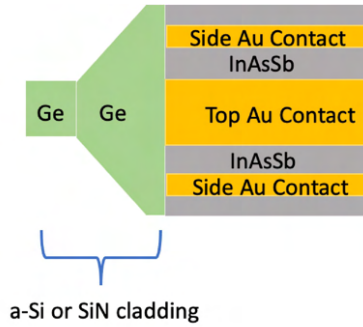


Figure 2-2: Electron band diagram for $\text{InAs}_{0.85}\text{Sb}_{0.15}$ heterojunction design, provided by Kazumi Wada.



(a) Cross Section with Metal Contacts Included



(b) Bird's Eye View

Figure 2-3: $\text{InAs}_{0.85}\text{Sb}_{0.15}$ homojunction device structure used in MWIR photodetector simulations. Metal contacts were only included in the FDTD simulation.

sweep key parameters, including the widths of the Ge waveguide and the $\text{InAs}_{0.85}\text{Sb}_{0.15}$ absorbing region. The effective confinement region shown in Figure 2-4 was defined as the portion of the absorbing region that would have an E field and would therefore contribute most to photocurrent. Ge waveguide width was evaluated between 3 and 10 μm , and the $\text{InAs}_{0.85}\text{Sb}_{0.15}$ pn region width from 1 to 8 μm . Simulations were performed at each 1 μm increment, with the obvious constraint that the Ge waveguide needed to be wider than the $\text{InAs}_{0.85}\text{Sb}_{0.15}$ pn region. As shown in Figure 2-5a and Figure 2-5b, both absorption and effective confinement increased when the $\text{InAs}_{0.85}\text{Sb}_{0.15}$ pn region was as wide as possible.

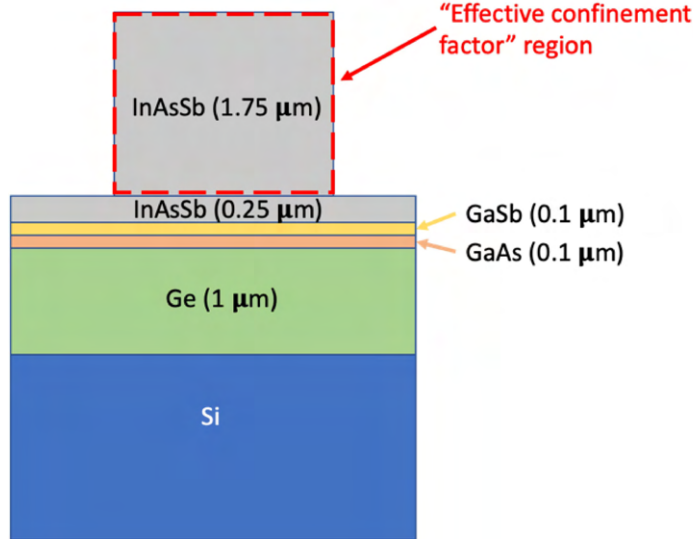
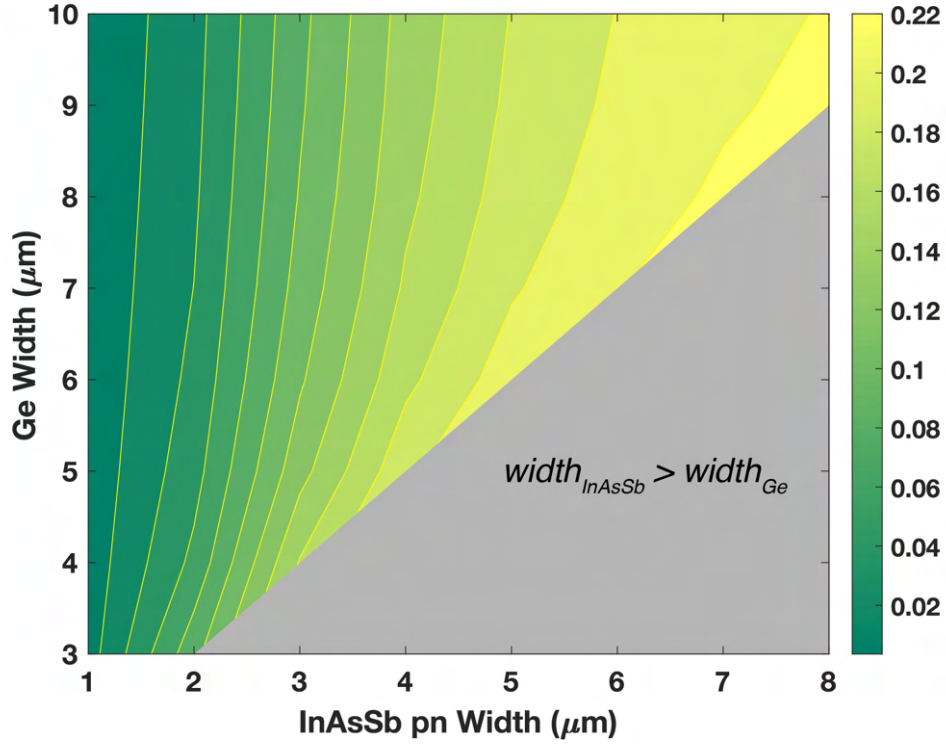
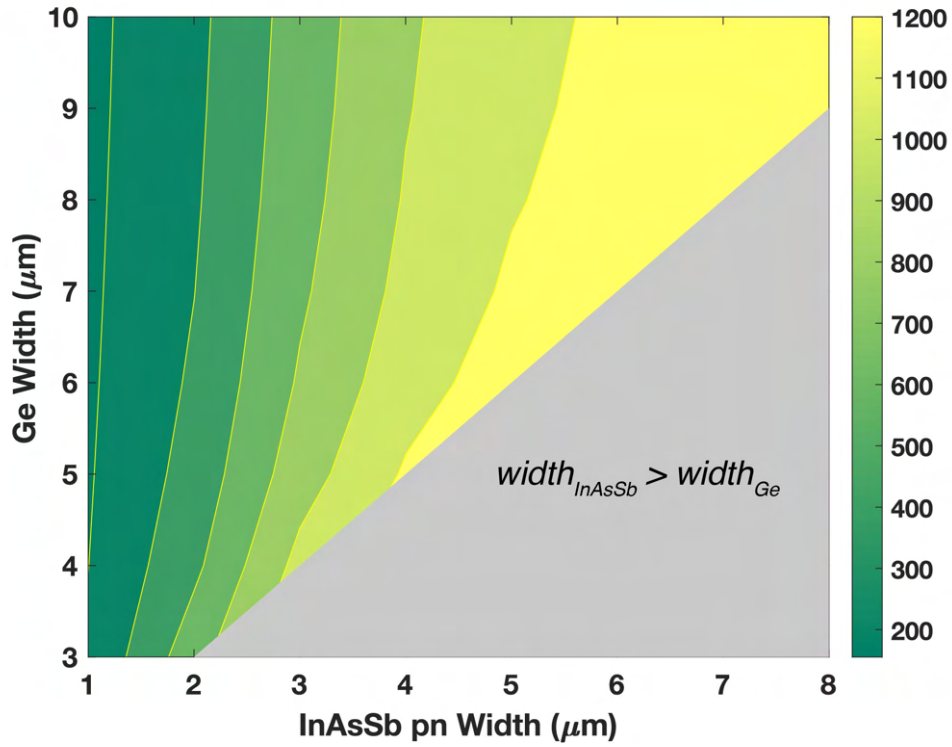


Figure 2-4: Cross section of $\text{InAs}_{0.85}\text{Sb}_{0.15}$ photodetector used in FDE sweeps, with effective confinement region defined.

For ease of fabrication, the total height of the $\text{InAs}_{0.85}\text{Sb}_{0.15}$ was limited to 2 μm , but the heights of the n+ region were also swept and evaluated against the same parameters. Increasing the total height clearly improved absorption as expected in, but maximizing the effective confinement also required that the $\text{InAs}_{0.85}\text{Sb}_{0.15}$ n+ region be minimized. Figure 2-6a and Figure 2-6b were checked for co-dependencies amongst the various parameters, confirming the validity of the individual parameter sweep results.

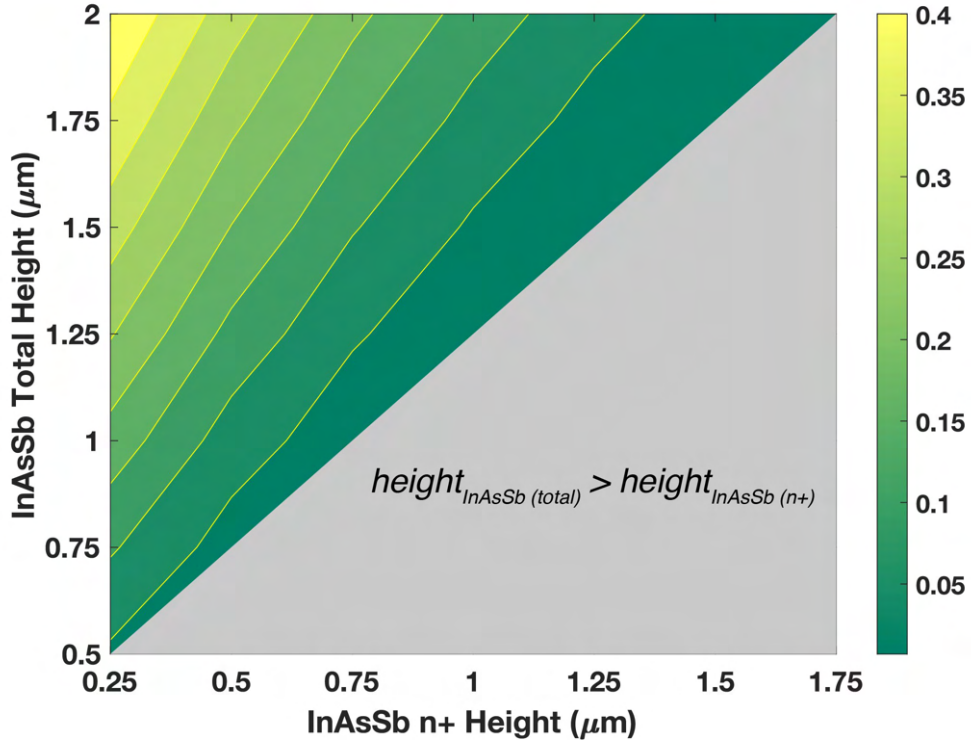


(a) Effective confinement factor of mode for various $\text{InAs}_{0.85}\text{Sb}_{0.15}$ photodetector widths.

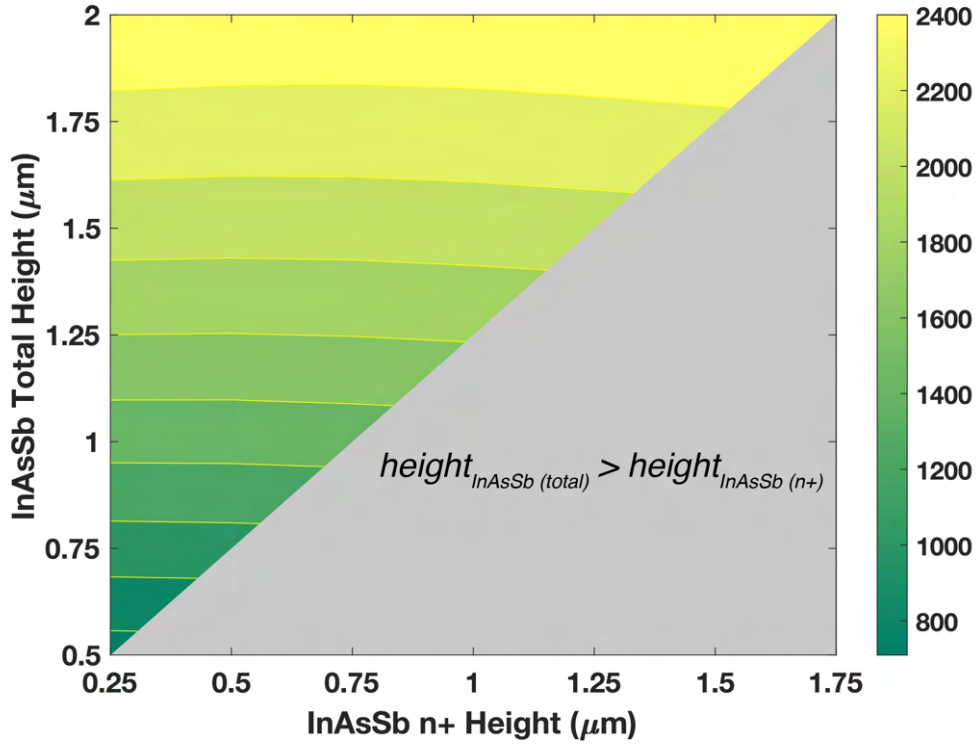


(b) Absorption (dB/cm) of mode for various $\text{InAs}_{0.85}\text{Sb}_{0.15}$ photodetector widths.

Figure 2-5: Simulation results from FDE Width Sweep.



(a) Effective confinement factor of mode for various $\text{InAs}_x\text{Sb}_{1-x}$ photodetector heights.



(b) Absorption (dB/cm) of mode for various $\text{InAs}_x\text{Sb}_{1-x}$ photodetector heights.

Figure 2-6: Simulation results from FDE Height Sweep.

The Ge taper length was simulated in EME solver (EigenMode Expansion) with $3\text{ }\mu\text{m}$ SiN cladding, a $2.6\text{ }\mu\text{m}$ input and $9\text{ }\mu\text{m}$ output waveguide. It was shown that $200\text{ }\mu\text{m}$ length guaranteed high transmission, and with enough footprint to accommodate such long lengths, no further optimization was necessary.

These photodetector dimensions were incorporated into an FDTD simulation (finite difference time domain), starting from the Ge output waveguide and ending $20\text{ }\mu\text{m}$ into the $\text{InAs}_{0.85}\text{Sb}_{0.15}$ photodetector region.

The E field from the source mode in the Ge waveguide manifested as several different modes in the photodetector region, with only 0.04% back reflection at the interface. The mode that was most confined in $\text{InAs}_{0.85}\text{Sb}_{0.15}$ had the highest absorption of 3353 dB/cm, with the other two modes at 786 dB/cm and 523 dB/cm.

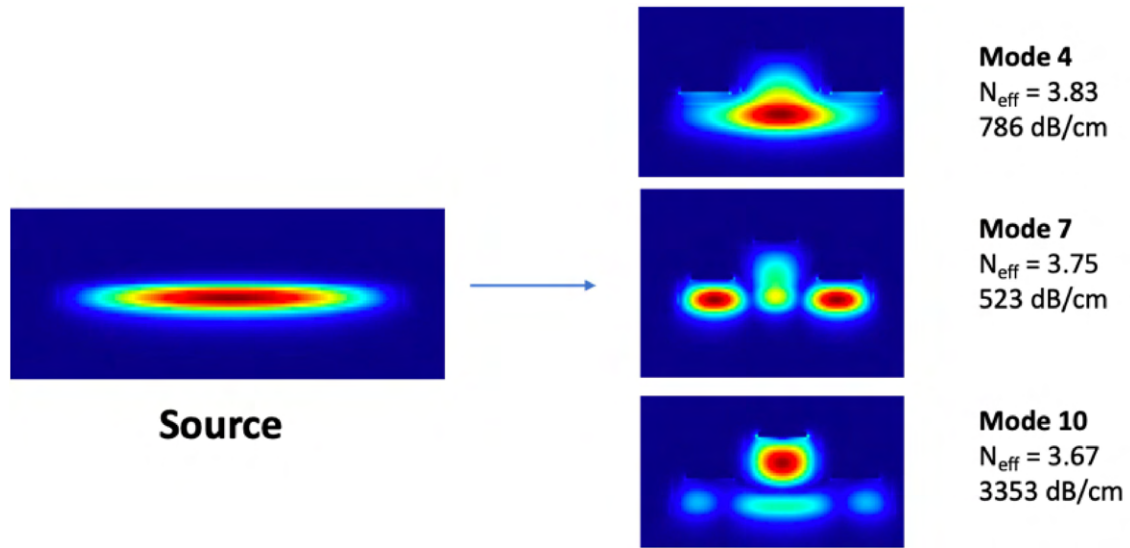


Figure 2-7: FDTD simulation of excited modes and their respective E fields in $\text{InAs}_{0.85}\text{Sb}_{0.15}$ photodetector, from source mode in the Ge waveguide.

The propagation of each mode in the photodetector region is shown [Figure 2-8](#). Evaluating the power fraction relative to the source, Mode 10 was absorbed quickly as expected, while the dominant Mode 4 dropped steadily, and Mode 7 remained essentially unchanged. The total forward power is also shown as dropping to 70% of its original power after a $20\text{ }\mu\text{m}$ photodetector length.

The slightly lower refractive index for $\text{InAs}_{0.85}\text{Sb}_{0.15}$ relative to Ge promotes mode propagation in the waveguide instead of the photodetector region. To improve power

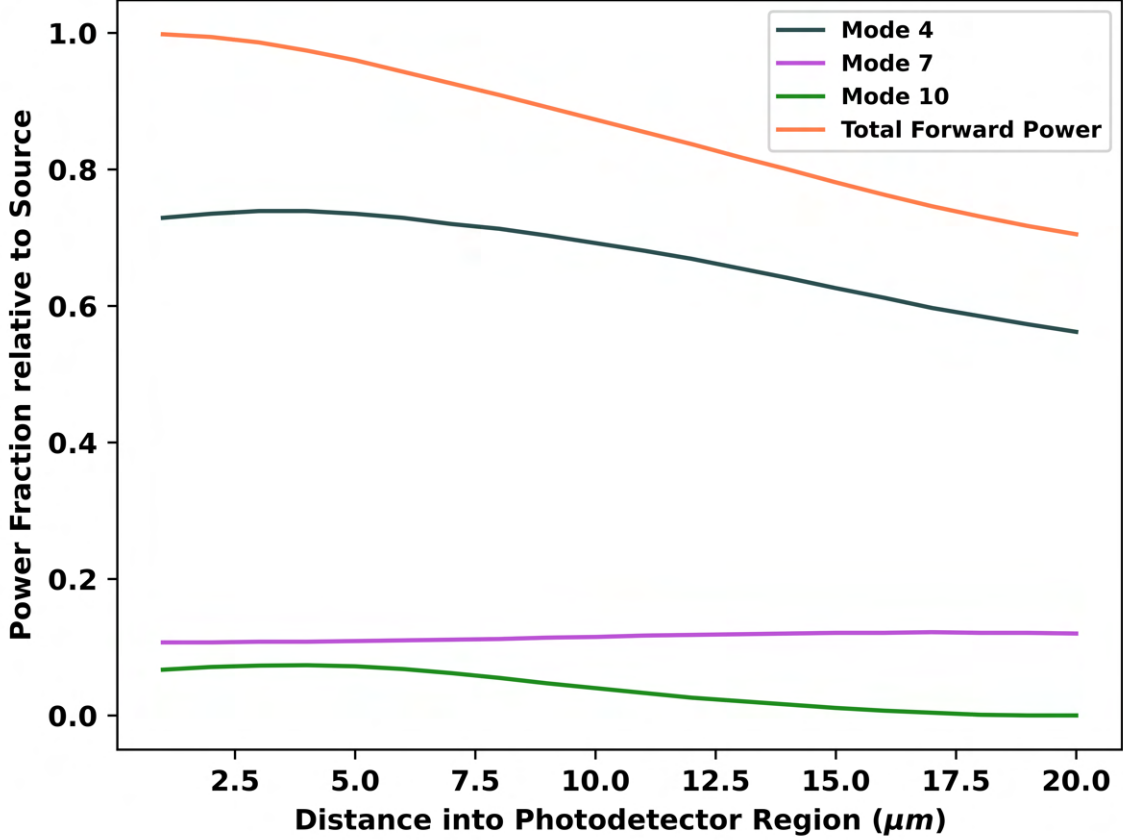


Figure 2-8: FDTD simulation of mode propagation within the $\text{InAs}_{0.85}\text{Sb}_{0.15}$ photodetector region.

coupling into the photodetector as well as mode confinement, the possibility of etching the Ge waveguide within the detector region was considered. An FDE sweep for the heterojunction design was performed for a $3\ \mu\text{m}$ wide, $1\ \mu\text{m}$ tall Ge waveguide.

In the detector region, the Ge thickness was swept from $0.25\ \mu\text{m}$ to $1.0\ \mu\text{m}$, while considering an $\text{InAs}_{0.85}\text{Sb}_{0.15}$ thickness of $0.25\ \mu\text{m}$ to $2.0\ \mu\text{m}$. Additional dimensions are shown in [Figure 2-9](#) and [Figure 2-10](#).

Although a thicker $\text{InAs}_{0.85}\text{Sb}_{0.15}$ layer improved absorption, the mode coupling efficiency from the input to the output tended to decrease. Thus, a new figure of merit was considered by multiplying these two parameters. It was demonstrated in [Figure 2-11](#) that for this alternative heterojunction design, etching the Ge waveguide as much as possible was highly beneficial, while keeping the $\text{InAs}_{0.85}\text{Sb}_{0.15}$ thickness to a moderate height of just over $1\ \mu\text{m}$ promoted both good mode confinement in the



Figure 2-9: Cross section of etched Ge waveguide in $\text{InAs}_{0.85}\text{Sb}_{0.15}$ photodetector region.

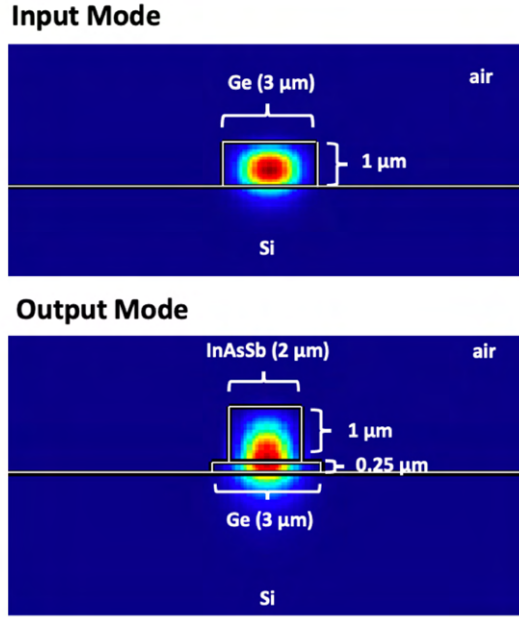


Figure 2-10: E field profiles of input mode for Ge waveguide and output mode in $\text{InAs}_{0.85}\text{Sb}_{0.15}$ photodetector region.

absorbing region and power coupling across the interface. The easiest to fabricate scenario, with the Ge waveguide completely unetched is also marked explicitly in [Figure 2-11](#), and demonstrates that reducing the Ge thickness in the photodetector region improves performance by a factor of 3x or more.

In summary, simulations were performed in Lumerical to optimize a MWIR photodetector that incorporated an $\text{InAs}_{0.85}\text{Sb}_{0.15}$ homojunction and was straightforward to fabricate, assuming successful epitaxy growths. The proposed photodetector design had the following dimensions: constant Ge waveguide thickness of 1 μm throughout, Ge waveguide and $\text{InAs}_{0.85}\text{Sb}_{0.15}$ n+ width = 10 μm , $\text{InAs}_{0.85}\text{Sb}_{0.15}$ pn width = 7 μm ,

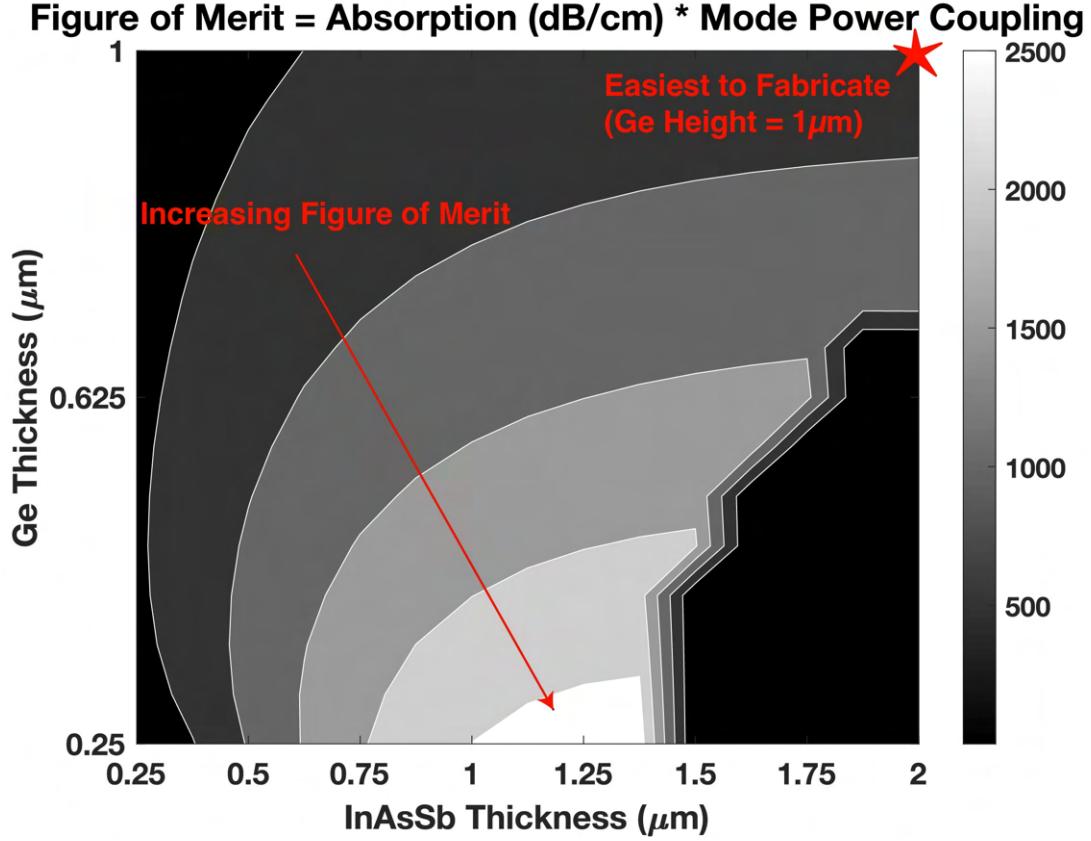


Figure 2-11: Figure of merit for integrated photodetector design with etched Ge in the $\text{InAs}_{0.85}\text{Sb}_{0.15}$ region.

$\text{InAs}_{0.85}\text{Sb}_{0.15}$ n+ height = $0.25\mu\text{m}$, and $\text{InAs}_{0.85}\text{Sb}_{0.15}$ pn height of $1.75\mu\text{m}$. The photodetector length promoted 30% light absorption after only $20\mu\text{m}$ propagation into the detection region, with an effective confinement of 45% and 2557 dB/cm absorption. The Ge input waveguide and standard taper were covered in $3\mu\text{m}$ SiN cladding, with the input waveguide width set to $2.6\mu\text{m}$ and a taper length of $200\mu\text{m}$.

A more difficult fabrication scenario, which etched the Ge waveguide within the photodetector region, was also proposed with initial demonstrations for an $\text{InAs}_{0.85}\text{Sb}_{0.15}$ heterojunction design. Instead of keeping the Ge thickness in the photodetector the same as in the waveguide region, the $1\mu\text{m}$ Ge waveguide was etched to a proposed thickness of $0.25\mu\text{m}$, allowing for a more efficient modal coupling at the interface and better confinement of the mode within the absorbing region.

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